Using deep sequencing to characterize the biophysical mechanism of a transcriptional regulatory sequence

Justin B. Kinney[a,b,1,2], Anand Murugan[a], Curtis G. Callan, Jr.[a,c,2], and Edward C. Cox[d]

[a]Department of Physics, Princeton University, Princeton, NJ 08544; [b]Lewis-Sigler Institute, Princeton University, Princeton, NJ 08544; [c]Princeton Center for Theoretical Science, Princeton University, Princeton, NJ 08544; and [d]Department of Molecular Biology, Princeton University, Princeton, NJ 08544

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Cells use protein-DNA and protein-protein interactions to regulate transcription. A biophysical understanding of this process has, however, been limited by the lack of methods for quantitatively characterizing the interactions that occur at specific promoters and enhancers in living cells. Here we show how such biophysical information can be revealed by a simple experiment in which a library of partially mutated regulatory sequences are partitioned according to their in vivo transcriptional activities and then sequenced en masse. Computational analysis of the sequence data produced by this experiment can provide precise quantitative information about how the regulatory proteins at a specific arrangement of binding sites work together to regulate transcription. This ability to reliably extract precise information about regulatory biophysics in the face of experimental noise is made possible by a recently identified relationship between likelihood and mutual information. Applying our experimental and computational techniques to the *Escherichia coli lac* promoter, we demonstrate the ability to identify regulatory protein binding sites de novo, determine the sequence-dependent binding energy of the proteins that bind these sites, and, importantly, measure the in vivo interaction energy between RNA polymerase and a DNA-bound transcription factor. Our approach provides a generally applicable method for characterizing the biophysical basis of transcriptional regulation by a specified regulatory sequence. The principles of our method can also be applied to a wide range of other problems in molecular biology.

gene regulation | lac promoter | mutual information | thermodynamic models | parallel tempering Monte Carlo

Cells regulate transcription primarily through the binding of proteins to DNA-binding sites within transcriptional regulatory sequences (TRSs). Understanding how TRSs use different arrangements of binding sites to encode regulatory programs remains a major challenge for molecular biology. High-throughput methods have spurred great progress in cataloging the genome-wide distribution of binding sites (1 and 2), and many techniques exist for characterizing the sequence specificity of individual regulatory proteins (3–6). However, determining how a specific TRS integrates information from multiple DNA-bound proteins still requires a laborious series of biochemical experiments that typically provide only qualitative information (reviewed in ref. 7).

The *E. coli lac* promoter (8 and 9) is one of the few TRSs whose function is well understood at the biophysical level (10 and 11). Kuhlman et al. (11) were the first to prove that a certain aspect of this system—the up-regulation of transcription by the protein CRP (12)—could be quantitatively explained by a simple energetic interaction between CRP and the σ^{70}-dependent RNA polymerase holoenzyme (henceforth RNAP). To do this, Kuhlman et al. measured transcriptional activity resulting from different in vivo concentrations of active CRP and showed that the resulting functional form of this activity was consistent with a simple thermodynamic model (13). By fitting the defining parameters of this model to their data, Kuhlman et al. were then able to measure the in vivo interaction energy between CRP and RNAP.

Despite its success, Kuhlman et al.'s approach is not feasible as a general method for studying TRSs. First, it requires quantitative control over the in vivo concentrations of all of the proteins that bind the TRS of interest. Secondly, TRSs typically contain multiple binding sites for each operative regulatory protein, making it difficult to determine the specific role of each site simply by varying the concentration of the regulator. This latter fact has created difficulty for similar studies of eukaryotic enhancers (14).

We hypothesized that measuring the activities of a large number of TRSs containing scattered point mutations could provide information similar to that produced by Kuhlman et al.'s method. This approach would be feasible for studying arbitrary TRSs, and would allow the effect of each individual binding site to be characterized. Because point mutations tend to preserve the spatial arrangement of binding sites, such measurements would allow one to interrogate the same protein-DNA complexes that allow the wild-type TRS to function. A similar approach had been tried by Schneider and Stormo in 1989 (15), but the recent advent of ultra-high-throughput sequencing, together with new techniques in machine learning (16) led us to believe that this approach could be much more powerful than had previously been realized.

In this article we report the application of this mutagenesis-based approach to the *E. coli lac* promoter. FACS (17) and 454 pyrosequencing (18) were used to characterize the activities of $\sim 200,000$ *lac* promoters mutagenized in a 75 bp region containing the CRP and RNAP binding sites (Fig. 1*A*). The resulting sequence data allowed us to identify these binding sites de novo, determine the sequence-dependent binding energy of both CRP and RNAP, and measure the in vivo interaction energy between these two proteins in their native DNA-bound configuration. We note that previous attempts to determine in vivo protein-protein interaction energies from sequence data (19 and 20) required unproven assumptions about how arbitrary arrangements of DNA-bound proteins interact; our approach does not.

In this way, we demonstrate how deep sequencing can be used to measure protein-DNA and protein-protein interaction energies in living cells. This ability should be useful for addressing many different questions in molecular biology.

Interrogating a TRS with Flow Cytometry and Deep Sequencing

We performed six experiments on region $[-75:-1]$ of the *E. coli lac* promoter (Fig. 1 *A* and *B*). These experiments, summarized in Table 1, differed in the positions within region $[-75:-1]$ that

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[1]To whom correspondence may be addressed. E-mail: jkinney@cshl.edu or ccallan@princeton.edu.

[2]Current address: Cold Spring Harbor Laboratory, Cold Spring Harbor, NY 11724.

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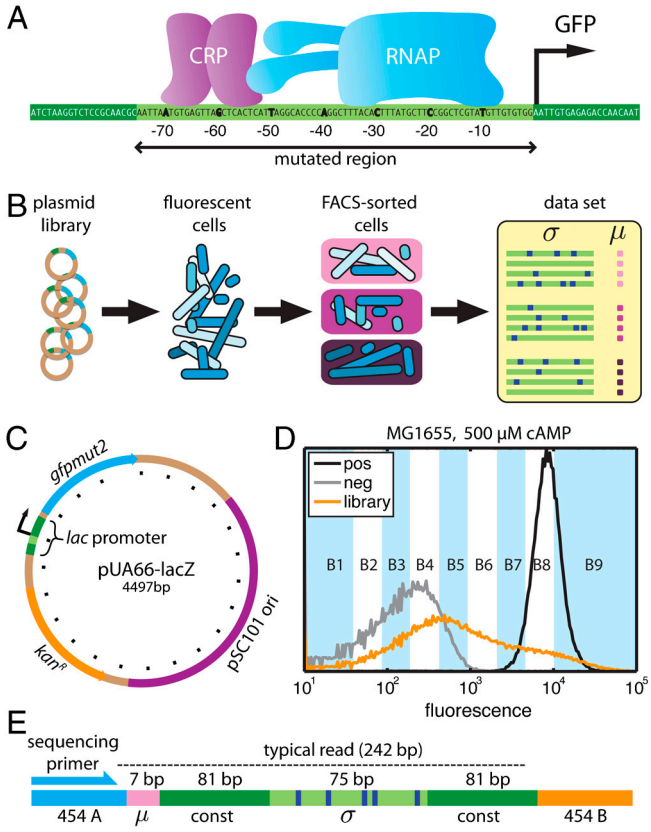

Fig. 1. Overview of the experiments. *A*) We used *lac* promoters mutagenized in region [−75: − 1] to drive the expression of GFP. *B*) Plasmids containing mutant *lac* promoters driving GFP expression were transformed into *E. coli*. Induced cells were then partitioned using FACS. Deep sequencing of the mutant promoters in each FACS batch yielded a long list of sequences σ with corresponding measurements μ. *C*) Plasmid pUA66-lacZ (21), a very-low-copy-number plasmid on which the wild-type *lac* promoter drives the expression of GFP; tick mark spacing is 200 bp. *D*) Fluorescence distributions of MG1655 cells containing the full-wt plasmid library (*orange*), the pUA66-lacZ plasmid (*black*), or a negative control plasmid pJK10 (*SI Appendix: Fig. S1*) in which region [−75: − 1] of the *lac* promoter was deleted (*gray*). In the full-wt experiment, batches B1–B9 received cells from the indicated fluorescence ranges, while batch B0 received cells randomly sampled from the initial library. *E*) Each PCR amplicon contained a 7 bp DNA barcode indicating the batch μ in which the sequence σ was found. 454 pyrosequencing (18) yielded reads of about 242 bp covering the indicated regions.

were mutagenized, the strain of *E. coli* used, and the physiological conditions under which *lac* promoter function was characterized.

The full-wt experiment used a library of reporter constructs, derived from pUA66-lacZ (Fig. 1*C*; ref. 21), in which region [−75: − 1] of the *lac* promoter was mutagenized at 12% per nucleotide, yielding 9 ± 3 substitution mutations per sequence. Wild-type *E. coli* (strain MG1655) were transformed with this plasmid library, after which GFP expression was induced during exponential growth in minimal media supplemented with glucose, cAMP, and 1 mM isopropyl β-D-1-thiogalactopyranoside (IPTG). FACS (17) was used to sort induced cells into 10 different batches, each cell according to its measured fluorescence (Fig. 1*D*). PCR was then used to tag mutant TRSs according to the batch in which each TRS was found (Fig. 1*E*). 454 pyrosequencing (18) of the resulting PCR amplicons yielded a list of 51,835 mutant TRSs and corresponding batches. The batch μ associated with each TRS σ serves as a noisy and qualitative measurement of that TRS's in vivo transcriptional activity. Experiments crp-wt and rnap-wt were performed the same way, but using mutant TRSs in which only the CRP or RNAP binding site was mutagenized (Table 1).

Table 1. Shown for each of our six experiments are the mutagenized region of the *lac* promoter, the per-position substitution rate, the *E. coli* strain used, the cAMP concentration used for induction, the number of batches into which cells were sorted, and the final number of filtered, nonredundant reads

Dataset	Mut. region	Mut. rate	Strain	cAMP (μM)	No. μ	No. reads
full-wt	[−75: − 1]	12%	MG1655	500	10	51,835
crp-wt	[−74: − 49]	24%	MG1655	500	10	46,986
rnap-wt	[−39: − 4]	15%	MG1655	500	10	45,461
full-500	[−75: − 1]	12%	TK310	500	5	23,431
full-150	[−75: − 1]	12%	TK310	150	5	24,334
full-0	[−75: − 1]	12%	TK310	0	5	28,544

The full-500, full-150, and full-0 experiments were performed using the same plasmid library as in full-wt, but with transcriptional activity assayed in *E. coli* strain TK310 (11) grown in 500 μM, 150 μM, or 0 μM cAMP, respectively. Cells were also sorted into five batches instead of 10. Strain TK310 lacks adenlyate cyclase (*cyaA*, needed for cAMP synthesis) and phosphodiesterase (*cpdA*, which degrades cAMP) and is therefore unable to control its intracellular cAMP levels (11). Growing TK310 cells in media supplemented with different concentrations of cAMP thus allowed us to control the active intracellular concentration of CRP (just as Kuhlman et al. did). Importantly, though, the mechanism of *lac* promoter function is the same in both MG1655 and TK310 cells.

In total we obtained 220,591 mutant TRSs, with each TRS σ assigned a noisy, qualitative measurement μ of its transcriptional activity (Fig. 1*B*). These sequences σ and measurements μ comprise the only data used in the analysis that follows.

Information Footprints Reveal Functional Binding Sites

Our first goal was to identify nucleotide positions that affect expression, thereby locating all functional binding sites within the probed region of the *lac* promoter. For this purpose we used mutual information, a fundamental quantity from information theory that provides the most general measure of dependence between any two variables (22). For each nucleotide position i, we computed the number of sequences in each batch μ having each of the four possible bases b_i. We then used this information to compute the mutual information $I(b_i;\mu)$ between b_i and μ (see *SI Appendix*: *Computing mutual information* for details), thereby quantifying the effect of the base at position i on our measurements. Plotting the mutual information $I(b_i;\mu)$ for each position i produced what we call an "information footprint."

Fig. 2*A* shows the information footprint produced by the full-wt experiment. The known binding sites of CRP and RNAP are clearly visible and each has the expected bipartite structure: CRP binds DNA as a homodimer at [−72: − 51] with a 6 bp gap between its two DNA-binding domains, while RNAP binding to [−41: − 1] results from the recognition of separate sequence elements centered roughly on positions −10 and −35 (8 and 12).

The information values displayed in Fig. 2*A* are small, ranging from ∼0.05 bits down to values indistinguishable from zero. This is not surprising, since each $I(b_i;\mu)$ measures the effect on transcription of just one out of 75 positions. But because our dataset is large ($N = 51,856$), these information values are determined very precisely—typically to within $\sim 4 \times 10^{-4}$ bits. Fig. 2*A* thus displays meaningful information values ranging over two orders of magnitude. This high level of sensitivity can reveal aspects of sequence function not detectable by other methods, e.g., (23). For instance, small but significant effects on expression were observed between the −10 and −35 elements and between the CRP and RNAP sites, regions not previously thought to influence transcription at the *lac* promoter. Indeed, only the 10 positions [−75, −70, −48, −42, −41, −23, −22, −21, −19, −3] show an insignificant effect on expression (i.e., $P > 0.05$).

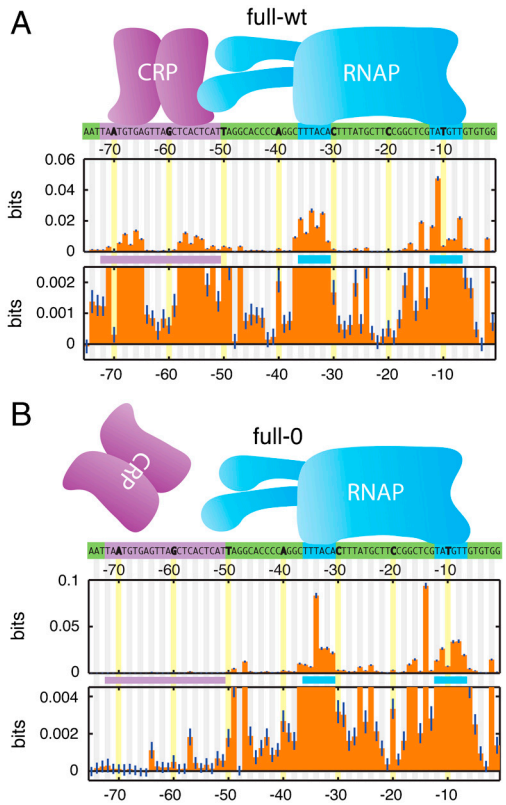

Fig. 2. Information footprints. *A*) Footprint from full-wt data, aligned with known protein-DNA contact positions (*highlighted*). The *lower plot* is a 20X magnification of the *upper plot*. Error bars (*dark blue lines*) indicate uncertainties due to finite sample effects (*SI Appendix*: *Computing mutual information*). *B*) Footprint from the full-0 experiment, in which intracellular CRP was inactive. *SI Appendix*: *Fig. S3* shows information footprints from all six experiments.

Information footprints can further allow one to determine which of the identified sites are involved in the response to a specific biochemical signal or genetic perturbation. Fig. 2*B* shows the information footprint from the full-0 experiment, in which TK310 cells were induced in the absence of cAMP, thereby inactivating intracellular CRP. The lack of active CRP is reflected in the near-total loss of information at positions $[-75:-51]$: information values at all but three positions within this 25 bp site differ insignificantly from zero (i.e., $P > 0.05$). An intermediate reduction in information occurs in the full-150 footprint, while the full-500 footprint closely resembles that of full-wt (*SI Appendix*: *Fig. S3*). We note that the small but significant information values at positions $[-64, -57, -52]$ in the full-0 footprint might result from interactions between DNA and the α subunits of RNAP (24), or from alternative RNAP binding sites (25).

An important caveat to this analysis is that the quantitative features of an information footprint ultimately depend on the details of one's experiment, including the level of mutagenesis used in the initial TRS library and the specific fluorescence gates used for sorting cells. So while qualitative differences between footprints from different experiments can be revealing, it is difficult to draw conclusions from more subtle quantitative differences, such as the different shapes of the RNAP footprint in Fig. 2 *A* and *B*. But as we show in the next section, explicit biophysical models can be fit to data in a way that does not depend on such experimental details. Quantitative differences between models inferred from different experiments can, as a result, be revealing about underlying molecular mechanisms.

Model Fitting in the Presence of Uncharacterized Noise

Next we used our data to infer quantitative models for how in vivo protein-DNA and protein-protein interactions modulate transcription. By "model" we mean a mathematical function that assigns to each sequence σ a predicted value x for some quantity of interest, such as the DNA-binding energy of a regulatory protein, or the rate of transcription resulting from the interactions between multiple proteins. To infer a given model, we first assumed a specific mathematical formula for the model, then determined the values of model parameters by matching the sequence-dependent predictions x to our sequence-specific measurements μ.

Such model fitting could be done in the standard Bayesian way if we knew the experimental "error model" $p(\mu|x)$—the probability of obtaining a measurement μ given an underlying quantity of interest x (such as binding energy or transcription rate). But in our experiments it was virtually impossible to accurately determine $p(\mu|x)$ a priori. Many difficult-to-characterize noise processes, including stochastic transcription, variations in cell size, and noise in FACS measurements, contribute to the quantitative form of $p(\mu|x)$. Also, in the case where x represents the DNA-binding energy of a protein, we do not know a priori how the binding of that protein affects transcription; indeed, this is something we want to learn from the data. Kinney et al. (16) overcame this problem by computing likelihood in the presence of an explicitly uncertain error model. They showed that regardless of what $p(\mu|x)$ actually is, the likelihood of a candidate model is well approximated by

$$p(\text{data}|\text{model}) = \text{const} \times 2^{NI(x;\mu)} \qquad \textbf{[1]}$$

in the limit where the number N of independently measured sequences is sufficiently large. Here $I(x;\mu)$ is the "predictive information" of the model—the mutual information between model predictions x and measurements μ. Although Kinney et al. focused on the analysis of microarray data, Eq. **1** is applicable to any dataset consisting of a large number of sequences and corresponding measurements. Kinney et al.'s approach therefore provides a practical substitute for standard likelihood-based inference when the experimental error model is either difficult to characterize or is unknowable a priori (see *SI Appendix*: *Statistical inference using mutual information* for more discussion).

In the analysis that follows, we used a custom parallel tempering Monte Carlo algorithm to sample model parameters according to the right-hand-side of Eq. **1** (see *SI Appendix*: *Parallel tempering Monte Carlo sampling of model parameters*). This allowed us to determine not just the best values for model parameters, i.e., which values maximize predictive information $I(x;\mu)$, but also the uncertainty in each parameter due to finite data. Because $N > 10^4$ for all of the experiments described in this article, even changes as small as 10^{-4} bits in the value of $I(x;\mu)$ led to substantial changes in model likelihood. The large amount of data produced by our experiments thus allowed us to determine precise quantitative values for model parameters even though our measurements were noisy and qualitative.

in Vivo Interaction Energies from Sequence Data

Having located the binding sites of both CRP and RNAP de novo using information footprints, we sought an explicit model for each protein's sequence-dependent binding energy. For this we used "energy matrix" models: each base within a protein's binding site was assumed to contribute additively to the overall binding energy. These simple models have been shown to accurately describe a number of transcription factors, (e.g., refs. 26 and 27), though there are known exceptions (28 and 29).

We fit an energy matrix for CRP to positions $[-74:-49]$ using the full-wt dataset. Energy matrix elements were sampled, according to Eq. **1**, using the predictive information $I(\varepsilon_c;\mu)$ where ε_c is CRP's predicted binding energy. The resulting optimal

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matrix is shown in Fig. 3*A*. We similarly fit an energy matrix to positions [−41 : − 1] to model RNAP's binding energy ε_r (Fig. 3*B*). CRP and RNAP energy matrices were also inferred from our five other datasets (*SI Appendix*: *Fig. S4A*). We note that at this stage of our analysis we were able to determine each matrix only up to an unknown multiplicative constant, not in physical units such as kcal/mol (see *SI Appendix*: *Maximizing mutual information leaves some model parameters undetermined*).

Unlike information footprints, these energy matrices are meant to capture intrinsic properties of the regulatory proteins, properties that should not depend on specific ways cells were sorted or on the level of mutagenesis used in the TRS library. The optimal matrices inferred from our six experiments (4 matrices for CRP, 5 for RNAP) are nearly identical, supporting this interpretation: CRP matrix elements derived from different experiments correlate by >95% (*SI Appendix*: *Fig. S4B*), while RNAP matrix elements exhibit >92% correlation (*SI Appendix*: *Fig. S4C*). Furthermore, each of these matrix models performs better on every one of our datasets than do any of the models for either CRP or RNAP currently in the literature (with two minor exceptions; see *SI Appendix*: *Model comparison to literature*). We find this level of quantitative agreement between experiments remarkable, considering that our six experiments used different promoter libraries, different *E. coli* strains (MG1655 or TK310), different inducing conditions (500 μM, 150 μM, or 0 μM cAMP), and different fluorescence gates for sorting cells. This close agreement in the face of important experimental differences attests to both the usefulness and correctness of Eq. **1**.

Our inferred CRP and RNAP energy matrices recapitulate much of what is known about the sequence specificities of these two proteins. The known consensus sequences—TGTGA$(N)_6$TCACA for CRP (3) and TTGACA$(N)_{18}$TATAAT for RNAP (30)—exactly match the lowest energy sequences predicted by nearly every one of our matrix models. The one exception is the RNAP matrix fit to full-0 data, which predicts that TTGATA

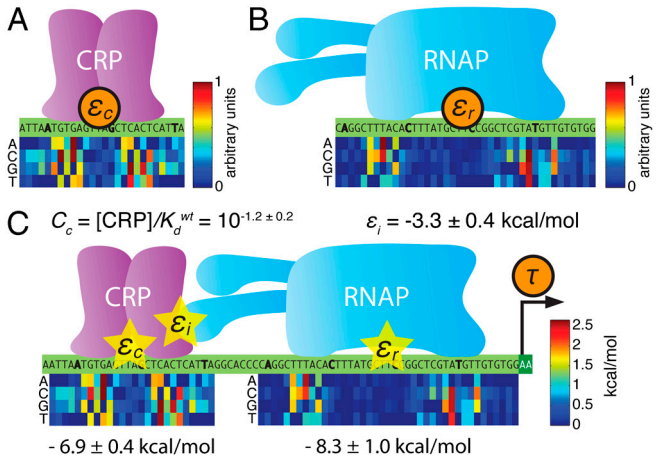

Fig. 3. Models fit to full-wt data. *A*) The CRP energy matrix fit to [−75 : − 49] by maximizing $I(\varepsilon_c;\mu)$ on full-wt data. *B*) The RNAP energy matrix fit to [−41 : − 1] by maximizing $I(\varepsilon_r;\mu)$ on full-wt data. In *A* and *B*, each matrix column lists the energy contributions of the four possible bases at the aligned position within the site. Matrix elements range from 0 to 1 (in arbitrary units) with the lowest element in each column set to zero by convention. *SI Appendix*: *Fig. S4* shows the CRP and RNAP matrices derived from all six of our datasets. *C*) The thermodynamic model for τ inferred using $I(\tau;\mu)$ in Eq. **1**. Optimal CRP and RNAP energy matrices are shown with elements expressed in kcal/mol (1 kcal/mol = $1.62k_bT$ at $T = 310$ °K). It is useful to define each wild-type *lac* promoter site as having zero energy. We therefore add an energy shift, shown *below* each matrix, when computing ε_c and ε_r. Doing this means that C_c represents the intracellular CRP concentration in units of the dissociation constant of the wild-type (zero energy) site. Values quoted for ε_i and C_c are mean ± rmsd values determined from the parameter ensembles sampled using parallel tempering Monte Carlo.

will have slightly lower energy than TTGACA in the −35 region. We note that every one of our RNAP matrices also predicts that having a "G" at position [−14] increases RNAP binding strength. In the literature this G is said to create an "extended -10 promoter," and such promoters are known to have increased transcriptional activity. Our CRP matrices are also in qualitative agreement with previous in vitro measurements (31), though there are some quantitative discrepancies.

Next we sought a quantitative understanding of how the interaction of CRP with RNAP affects transcription. Kuhlman et al. (11) previously showed that a simple biophysical model based on equilibrium statistical mechanics (reviewed in ref. 13) accounted well for the effect of cAMP on *lacZ* expression in TK310 cells. We hypothesized that using energy matrices to describe the binding energies of CRP and RNAP within Kuhlman et al.'s model, then fitting all model parameters to our data de novo, would allow us to recover Kuhlman et al.'s results, including their measurement of the interaction energy between CRP and RNAP.

Following Kuhlman et al., we assumed that the rate of transcription τ at the *lac* promoter is proportional to the occupancy of RNAP at its binding site in thermal equilibrium. This model is quantitatively expressed as

$$\tau = \tau_{\max}\frac{C_r e^{-\varepsilon_r/RT} + C_c C_r e^{-(\varepsilon_c+\varepsilon_r+\varepsilon_i)/RT}}{1 + C_c e^{-\varepsilon_c/RT} + C_r e^{-\varepsilon_r/RT} + C_c C_r e^{-(\varepsilon_c+\varepsilon_r+\varepsilon_i)/RT}}, \quad [2]$$

where RNAP occupancy is given by the sum of Boltzmann weights corresponding to physical states in which RNAP is bound, divided by the sum of weights for all possible states of the system. These Boltzmann weights depend on (*i*) the CRP and RNAP binding energies ε_c and ε_r, which we express in kcal/mol and normalize to be zero at each wild-type *lac* promoter site, (*ii*) the concentrations C_c and C_r of CRP and RNAP, expressed in units of each wild-type site's dissociation constant and (*iii*) the CRP-RNAP interaction energy ε_i, expressed in kcal/mol. $\tau_{\max}$ is the transcription rate resulting from full RNAP occupancy. $R = 1.98 \times 10^{-3}$ kcal/mol °K is the gas constant and $T = 310$ °K (37 °C) is the temperature at which cells were induced.

Using $I(\tau;\mu)$ evaluated on full-wt data, we fit all of the parameters defining τ, including ε_i, C_c, and the elements of the energy matrices used to compute ε_c and ε_r. Doing so we inferred a CRP-RNAP interaction energy $\varepsilon_i = -3.26 \pm 0.41$ kcal/mol. This value is consistent with Kuhlman et al.'s measurement of −3.4 kcal/mol (11), thus demonstrating that DNA sequence data can be used to measure the in vivo interaction energy between two proteins. This procedure also yielded an in vivo CRP concentration of $C_c = [\mathrm{CRP}]/K_d^{\mathrm{wt}} = 10^{-1.2\pm0.2}$. Fig. 3*C* shows these values for ε_i and C_c, as well as the optimal energy matrices for ε_c and ε_r inferred by fitting τ. These matrices closely resemble those in Fig. 3 *A* and *B*, but, unlike the matrices we inferred by separately fitting ε_c and ε_r, their elements are determined explicitly in physical units of kcal/mol. We note that fitting τ to full-wt data provided no information about the value of either $\tau_{\max}$ or C_r (see *SI Appendix*: *Maximizing mutual information leaves some model parameters undetermined*).

Testing Biochemical Mechanisms by Fitting a Single Model to Multiple Datasets

cAMP is known to alter *lac* promoter activity by affecting CRP's ability to bind DNA, not CRP's interaction with RNAP. Both of these possibilities, though, are consistent with the information footprints shown in Fig. 2. By contrast, the former hypothesis predicts that the CRP concentration C_c in our model for τ should vary from experiment to experiment, while the latter predicts an experiment-dependent interaction energy ε_i. To further test the validity of our approach, we fit a single model for τ to all six of our datasets (see *SI Appendix*: *Fitting a model to multiple data sets*). This multidataset model employed a single CRP energy

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matrix and a single RNAP energy matrix, but allowed for dataset-specific values for both ε_i and C_c.

The experiment-specific values we inferred for ε_i and C_c are shown in Fig. 4 *A* and *B*. The ε_i values determined for all six experiments are mutually consistent ($P > 0.05$, χ^2 test), whereas the six C_c values differ very significantly: C_c values for experiments full-0 and full-150, which were performed in reduced concentrations of cAMP, were found to be much lower than the C_c values determined for the other four experiments. These results verify the well known fact that cAMP affects the concentration of active CRP, not the strength of CRP's interaction with RNAP (8 and 12).

Repeating this inference using only one ε_i value for all six experiments yielded the final inferred model of this article (Fig. 4 and *SI Appendix: Fig. S5*). The CRP-RNAP interaction energy inferred from all six datasets is $\varepsilon_i = -2.82 \pm 0.13$ kcal/mol. This differs from Kuhlman et al.'s measurement, but only by ∼20%. Also, the ratio of CRP concentrations inferred for full-150 and full-500 ($C_c^{\text{full}-150}/C_c^{\text{full}-500} = 10^{-0.56\pm0.02}$) closely agrees with the ratio of cAMP (150 μM/500 μM $= 10^{-0.52}$) used in these two experiments. This agreement is consistent with Kuhlman et al.'s observation that, for TK310 cells, the concentration of active CRP in vivo is proportional to the exogenous concentration of cAMP (11). Indeed, even the expected trace amount of cAMP present in the full-0 induction media (∼50 nM, a result of carry-over from the starter culture inoculum) is fully consistent with the much reduced yet significantly nonzero value for $C_c^{\text{full}-0}/C_c^{\text{full}-500} = 10^{-3.7\pm0.6}$.

We note, however, that there are also puzzling quantitative oddities in our results. Our CRP concentration ratios are in good agreement with expectations, but the absolute values we inferred ($C_c^{\text{full}-500} = 10^{-1.13\pm0.06}$ and $C_c^{\text{full}-150} = 10^{-1.70\pm0.07}$) are about 20-fold lower than the corresponding values claimed by Kuhlman et al. (*SI Appendix*: *Model comparison to literature*). Also, in the six-ε_i fit, the average ε_i is -2.34 ± 0.09 kcal/mol, substantially less than the -2.82 ± 0.13 kcal/mol we inferred using a common ε_i for all six experiments. Finally, all of our inferred CRP energy matrices are asymmetric and predict that CRP binds in the energetically unfavorable orientation at the wild-type CRP site. We think these strange results probably result from our models for ε_c, ε_r, and τ being too simplistic. The issue of model selection is complicated, however, and is beyond the scope of this article.

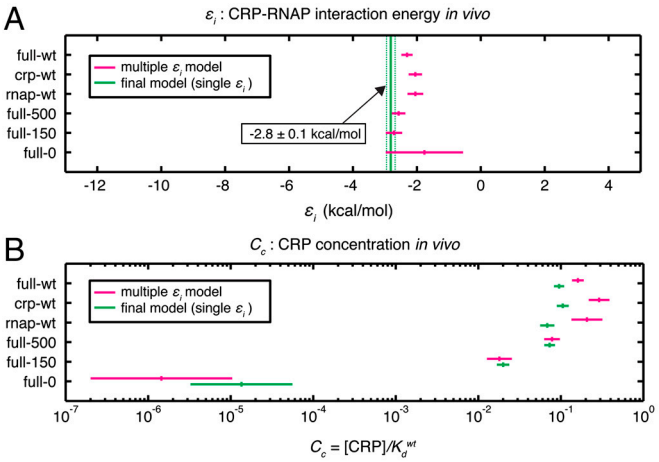

Fig. 4. Parameters fit to all six datasets. *A*) CRP-RNAP interaction energies ε_i (mean ± rmsd) inferred by fitting τ to all six datasets, using either data-set-specific values for ε_i (*magenta*) or a single ε_i for all six datasets (*green*). *B*) CRP concentrations C_c inferred for these same multidataset models. *SI Appendix: Fig. S5* shows full ensemble distributions for ε_i and the six C_c parameters of the final model, together with mean and rmsd values for all the CRP and RNAP matrix elements.

Model Validation Using Predictive Information

Predictive information can be used to determine how well a proposed thermodynamic model integrates binding energies into a single transcriptional output. The predictive information $I(\tau;\mu)$ quantifies how well the biophysical model in Eq. **2** accounts for our measurements μ. However, we can also directly compute the predictive information $I(\varepsilon_c, \varepsilon_r; \mu)$ of the pair of binding energies, ε_c and ε_r, without any model for τ (*SI Appendix*: *Computing mutual information*). The mere fact that τ is a function of ε_c and ε_r (in Eq. **2**) means that $I(\tau;\mu) \leq I(\varepsilon_c, \varepsilon_r; \mu)$, i.e., the predictive information of τ is bounded above by the predictive information of the pair of energies ε_c and ε_r. This inequality is a direct consequence of the Data Processing Inequality, a basic result in information theory (22). In our case, equality between $I(\tau;\mu)$ and $I(\varepsilon_c, \varepsilon_r; \mu)$ can be achieved only if τ preserves all of the transcriptionally relevant information encoded in the predicted values for ε_c and ε_r. We emphasize that there is no a priori guarantee that any quantity τ can do this, let alone a quantity derived from a simple biophysical model.

Remarkably we find (on full-wt data) that $I(\tau;\mu) = 0.732 \pm 0.007$ bits, which is identical within error bars to $I(\varepsilon_c, \varepsilon_r; \mu) = 0.732 \pm 0.006$ bits. We believe this equality is an important validation of the specific thermodynamic formula used to represent τ (Eq. **2**). This agreement also argues that our energy matrix models for ε_c and ε_r provide a valid representation of physical binding energy; if they did not, their use in the Boltzmann exponents in Eq. **2** would be unlikely to yield sensible results. We note that a simpler model in which transcription depends only linearly on ε_c and ε_r—which would be appropriate if CRP and RNAP bound DNA only as a complex—achieves only $I(\varepsilon_c + \varepsilon_r; \mu) = 0.647 \pm 0.005$ bits when all parameters are fit ab initio to full-wt data. This value is significantly less than $I(\varepsilon_c, \varepsilon_r; \mu)$. Our specific thermodynamic model for τ, with its physically motivated functional form, therefore provides a marked improvement over the more naive linear model.

Discussion

The approach we present here can be applied to a wide variety of TRSs in a number of different organisms. No prior knowledge of a TRS's sequence architecture is needed. All that is required is that (*i*) the TRS of interest function on a reporter construct and (*ii*) a large library of reporter constructs be introduced into cells so that each cell receives a single mutant TRS. After the activity-based partitioning and sequencing of mutant TRSs, information footprints can be used to identify all functionally relevant positions within the probed sequence. Information footprints from experiments performed in growth conditions or genetic backgrounds that are known to affect expression can further help one identify which "clumps" of informative positions correspond to discrete binding sites, as well as which of these sites are involved in transducing specific intracellular signals. Eq. **1** then allows one to infer mathematical models describing the sequence-dependent binding energy of each site's cognate protein. The same fitting procedure can also be used to build biophysical models of the in vivo interactions between multiple DNA-bound proteins. Such inference requires no quantitative model of experimental noise, thus allowing experiments performed in very different ways to produce nearly identical results.

GFP reporter plasmids have been constructed for almost all *E. coli* promoters (21). Starting from these plasmids and using our protocols, it should be possible to biophysically characterize the vast majority of promoters in *E. coli*. Similar experiments can likely be performed for most TRSs in yeast. While we demonstrated our technique on a TRS containing only two protein binding sites, the great sensitivity of this approach should allow the simultaneous effects of many DNA-binding proteins (including nucleosomal proteins) to be discerned from a single experiment. Ultimately, our method should be useful for characterizing the

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detailed functional architecture of TRSs active in a wide variety of culturable cells, including stem cells and cancer cell lines. The biophysical characterization of TRSs in living animals, however, will likely require significant changes to the experimental approach described here.

The underlying principles of our approach should also be useful for studying many different systems in which sequence-dependent interactions (e.g. protein-DNA, protein-RNA, protein-protein or protein-ligand interactions) play a central role in establishing some activity of interest. Experiments like ours can be performed both in vivo and in vitro; one simply needs to partition a large number of mutant sequences according to each one's activity. Our analysis method should then allow quantitative models of sequence-dependent function to be rigorously inferred from the resulting sequence data, regardless of how the activity-dependent partitioning of sequences is accomplished.

Experimental Procedures

See *SI Appendix*: *Supporting experimental procedures* for an expanded explanation of our experimental procedures.

Strains. *E. coli* strains MG1655 (wild type) and TK310 ($\Delta cyaA$ $\Delta cpdA$ $\Delta lacY$) were kindly provided by Thomas Kuhlman. Except where noted, TK310 cells were maintained in media supplemented with 500 μM cAMP in order to prevent *crp*-mediated suppression of $\Delta cyaA$ (32).

Library Construction. Mutant *lac* promoter libraries were synthesized using defined mixtures of nucleoside phoshporamidites. Plasmid libraries consisted of $\sim 2 \times 10^6$ independently cloned plasmids in which region $[-75: -1]$ was exactly replaced without the introduction of artificial restriction sites (*SI Appendix*: *Plasmid library construction*)

Sorting. Cells were grown in exponential phase for ≥10 generations, diluted into buffer, and stored on ice for 0–24 h prior to sorting. A BD Biosciences FACSVantage SE with DiVa was then used to sort 100,000 cells into each batch based solely on GFP fluorescence. Plating revealed ∼70, 000 viable cells per batch.

Amplicon Generation and Sequencing. Miniprepped plasmid from each FACS batch was used as template for amplicon-generating PCR. Two control sets of amplicons were also generated from pUA66-lacZ plasmid. The 47 resulting amplicon libraries (45 FACS batches + 2 wild-type controls) were collated and sequenced by Roche using the Genome Sequencer FLX platform. This yielded 448,416 sequences, 308,309 of which passed our quality filters. Unfortunately, an analysis of these sequences indicated a large postsort reduction in sequence diversity (*SI Appendix*: *Post-sort loss of library diversity*). To guarantee that each sequence was independently sorted, we discarded all but one copy of each sequence in each batch, leaving a total of 220,591 sequences across our six experiments.

Sequence Data and Analysis Results. Our 454 sequence data is available on the NCBI website under accession number SRA012345. Additional information, including processed sequence reads and inferred model parameters, is available at http://www.princeton.edu/~ccallan/sortseq09/.

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BIOPHYSICS AND COMPUTATIONAL BIOLOGY

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Using deep sequencing to characterize the biophysical mechanism of a transcriptional regulatory sequence: SI Appendix

Justin B. Kinney, Anand Murugan, Curtis G. Callan Jr., Edward C. Cox

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1. Supporting experimental procedures

1a. Plasmid library construction

Fig. S1 shows the *lac* promoter region of the plasmid pUA66-lacZ (panel A). Region [-75:-1] of this promoter is bounded on the left by the 4 bp sequence GCAA and on the right by the 4 bp sequence TGTG (brown). To facilitate the replacement of this region with mutant sequences without introducing artificial restriction sites into the *lac* promoter, we created plasmid pJK10 (Fig. S1B), in which the region between GCAA and TGTG was replaced with a polylinker sequence containing two outward-facing binding sites for the type IIs restriction enzyme BsmBI. Details on how pJK10 was constructed from pUA66-lacZ are provided in Kinney, 2008. BsmBI binds the 6 bp sequence CGTCTC (red) and cuts downstream of this sequence, leaving 5' overhangs at the bounding sequences GCAA and TGTG. To ease the preparation of plasmid libraries, we further inserted a DNA cassette containing the *ccdB* gene (see Hartley et al., 2000) into the NheI site of pJK10 (blue), yielding vector pJK14 (Fig. S1C). pJK14 was propagated in DB3.1 *E. coli* (Invitrogen), a strain immune to the CcdB toxin.

Single-stranded 119 bp DNA oligos containing mutant [-75:-1] regions were synthesized by IDT (Fig. S1D); defined mixtures of nucleoside phosphoramidites were used to give desired per-nucleotide mutation rates within specified variable regions (see Table 1, main text). Three oligo libraries were synthesized: iJK10-full (12% substitution rate in [-75:-1]), iJK10-crp (24% substitution rate in [-74:-49]), and iJK10-rnap (15% substitution rate in [-36:-4]). Inward-facing binding sites (green) for the type IIs restriction enzyme BsaI were positioned outside the 4 bp bounding sequences so that digestion with BsaI would again create 5' overhangs at GCAA and TGTG. Synthesized oligos were made double-stranded by primer annealing and Klenow extension.

To create plasmid libraries, pJK14 vector was digested with BsmBI (NEB) and PCR purified (Qiagen) after heat inactivation of this restriction enzyme. Double-stranded iJK10 inserts were separately digested with BsaI and PCR purified. To avoid dissociation of these double-stranded oligos, heat inactivation was not used. Digested pJK14 and iJK10 were ligated together at the GCAA and TGTG overhangs using T4 DNA ligase (Invitrogen), desalted (Atrazhev and Elliott, 1996), and electroporated into ElectroMax DH5α-E *E. coli* (Invitrogen). This yielded ~5×10^6 transformants per reaction. Using type IIs restriction enzymes in this way allowed exact replacement of region [-75:-1] without the introduction of artificial restriction sites (as described in Beck and Burtscher, 1994). Also, because *ccdB* is fatal when expressed in DH5α-E cells, cut pJK14 did not have to be gel purified before ligation with iJK10; a control ligation lacking iJK10 inserts reduced transformation efficiencies by a factor of ~10^3. Each plasmid library was miniprepped (Qiagen) and electroporated into electrocompetent MG1655 or TK310 E. coli (prepared using standard methods; Sambrook and Russel, 2001), yielding ~2×10^6 transformants per reaction.

1b. Media and growth rates

This study made frequent use of Vogel Bonner minimal media (Vogel and Bonner, 1956), both unsupplemented (VB buffer) and supplemented with 0.5% glucose and 50 ng/μl kanamycin (VB media). All growth was performed at 37 °C (310 °K). Prior to GFP induction, starter cultures were grown to saturation in VB media supplemented with 500 μM cAMP. Inductive growth took place in VB media supplemented with 1 mM IPTG and experiment-dependent concentrations of cAMP (Table 1, main text). Cells were diluted into VB buffer and placed on ice prior to FACS sorting. Cells were sorted into VB media containing 500 μM cAMP and immediately placed on ice. Sorted cells were then either plated on VB media plates (1% agar, 500 μM cAMP) or grown to saturation in VB media containing 500 μM cAMP prior to plasmid miniprep for amplicon PCR.

Doubling times for the various libraries during pre-FACS induction are shown in Table S1. Generation times were inferred from the growth curves in Fig. S2A-D. The lack of any noticeable growth rate difference between positive control cultures (carrying pUA66-lacZ) and negative control cultures (carrying pJK10), in any condition or cell type, shows that GFP expression did not have a significant effect on growth rate. By contrast, the amount of cAMP used to supplement TK310 cultures had a very large effect on growth rate.

Table S1: Doubling times during induction

Culture	Doubling time (min)	Culture	Doubling time (min)
full-wt	67.0	neg-500	61.8
crp-wt	66.4	full-150	68.6
rnap-wt	66.7	pos-150	68.9
pos-wt	67.2	neg-150	67.8
neg-wt	68.3	full-0	157
full-500	67.0	pos-0	154
pos-500	62.9	neg-0	156

1c. Amplicon generation

Table S2 shows the primers used for generating amplicons. Amplicons from each FACS batch were generated by PCR using the right primer aJK4.R and one of the 47 barcoded left primers aJK4.L.01 to aJK4.L.47. Green text indicates pUA66-lacZ binding sequences. 454 sequencing primer binding sites are shown in blue (primer A) and orange (primer B). Batch-specific barcodes are indicated by black text. These 7 bp barcodes do not contain homopolymer sequences and differ from one another in at least three positions.

Table S2: Primers used for generating amplicons

Batch	Primer	Sequence
all	aJK4.R	GCCTTGCCAGCCCGCTCAGGACGTTGTAAAACGACGGCCAGTGAATCCGT
crp-wt B0	aJK4.L.01	GCCTCCCTCGCGCCATCAGCGACGCTTAGGCGTATCACGAGGCCCTTTCGTCTTC
crp-wt B1	aJK4.L.02	GCCTCCCTCGCGCCATCAGATGTCTCTAGGCGTATCACGAGGCCCTTTCGTCTTC
crp-wt B2	aJK4.L.03	GCCTCCCTCGCGCCATCAGACGCACGTAGGCGTATCACGAGGCCCTTTCGTCTTC
crp-wt B3	aJK4.L.04	GCCTCCCTCGCGCCATCAGCATCGTGTAGGCGTATCACGAGGCCCTTTCGTCTTC
crp-wt B4	aJK4.L.05	GCCTCCCTCGCGCCATCAGACATGACTAGGCGTATCACGAGGCCCTTTCGTCTTC
crp-wt B5	aJK4.L.06	GCCTCCCTCGCGCCATCAGCGCTCTGTAGGCGTATCACGAGGCCCTTTCGTCTTC
crp-wt B6	aJK4.L.07	GCCTCCCTCGCGCCATCAGAGTGTCTTAGGCGTATCACGAGGCCCTTTCGTCTTC
crp-wt B7	aJK4.L.08	GCCTCCCTCGCGCCATCAGCGTACACTAGGCGTATCACGAGGCCCTTTCGTCTTC
crp-wt B8	aJK4.L.09	GCCTCCCTCGCGCCATCAGTAGCATGTAGGCGTATCACGAGGCCCTTTCGTCTTC
crp-wt B9	aJK4.L.10	GCCTCCCTCGCGCCATCAGCTAGATATAGGCGTATCACGAGGCCCTTTCGTCTTC
full-wt B0	aJK4.L.11	GCCTCCCTCGCGCCATCAGTCACGTGTAGGCGTATCACGAGGCCCTTTCGTCTTC
full-wt B1	aJK4.L.12	GCCTCCCTCGCGCCATCAGATGCTATTAGGCGTATCACGAGGCCCTTTCGTCTTC
full-wt B2	aJK4.L.13	GCCTCCCTCGCGCCATCAGAGATATGTAGGCGTATCACGAGGCCCTTTCGTCTTC
full-wt B3	aJK4.L.14	GCCTCCCTCGCGCCATCAGATATAGCTAGGCGTATCACGAGGCCCTTTCGTCTTC
full-wt B4	aJK4.L.15	GCCTCCCTCGCGCCATCAGCTGACGATAGGCGTATCACGAGGCCCTTTCGTCTTC
full-wt B5	aJK4.L.16	GCCTCCCTCGCGCCATCAGAGCTGTATAGGCGTATCACGAGGCCCTTTCGTCTTC
full-wt B6	aJK4.L.17	GCCTCCCTCGCGCCATCAGAGTGATATAGGCGTATCACGAGGCCCTTTCGTCTTC
full-wt B7	aJK4.L.18	GCCTCCCTCGCGCCATCAGCATGACATAGGCGTATCACGAGGCCCTTTCGTCTTC
full-wt B8	aJK4.L.19	GCCTCCCTCGCGCCATCAGTGTATGCTAGGCGTATCACGAGGCCCTTTCGTCTTC
full-wt B9	aJK4.L.20	GCCTCCCTCGCGCCATCAGTGTCAGTTAGGCGTATCACGAGGCCCTTTCGTCTTC
rnap-wt B0	aJK4.L.21	GCCTCCCTCGCGCCATCAGTGTACTGTAGGCGTATCACGAGGCCCTTTCGTCTTC
rnap-wt B1	aJK4.L.22	GCCTCCCTCGCGCCATCAGCTGTGCTTAGGCGTATCACGAGGCCCTTTCGTCTTC
rnap-wt B2	aJK4.L.23	GCCTCCCTCGCGCCATCAGTATCTACTAGGCGTATCACGAGGCCCTTTCGTCTTC
rnap-wt B3	aJK4.L.24	GCCTCCCTCGCGCCATCAGTCGTGTCTAGGCGTATCACGAGGCCCTTTCGTCTTC

rnap-wt B4	aJK4.L.25	GCCTCCCTCGCGCCATCAGACGTCGATAGGCGTATCACGAGGCCCTTTCGTCTTC
rnap-wt B5	aJK4.L.26	GCCTCCCTCGCGCCATCAGTGCTGAGTAGGCGTATCACGAGGCCCTTTCGTCTTC
rnap-wt B6	aJK4.L.27	GCCTCCCTCGCGCCATCAGTAGACGTTAGGCGTATCACGAGGCCCTTTCGTCTTC
rnap-wt B7	aJK4.L.28	GCCTCCCTCGCGCCATCAGCAGCAGATAGGCGTATCACGAGGCCCTTTCGTCTTC
rnap-wt B8	aJK4.L.29	GCCTCCCTCGCGCCATCAGTACATCTTAGGCGTATCACGAGGCCCTTTCGTCTTC
rnap-wt B9	aJK4.L.30	GCCTCCCTCGCGCCATCAGACAGCATTAGGCGTATCACGAGGCCCTTTCGTCTTC
full-500 B1	aJK4.L.31	GCCTCCCTCGCGCCATCAGAGAGCTCTAGGCGTATCACGAGGCCCTTTCGTCTTC
full-500 B2	aJK4.L.32	GCCTCCCTCGCGCCATCAGATACGCATAGGCGTATCACGAGGCCCTTTCGTCTTC
full-500 B3	aJK4.L.33	GCCTCCCTCGCGCCATCAGTCTGTCGTAGGCGTATCACGAGGCCCTTTCGTCTTC
full-500 B4	aJK4.L.34	GCCTCCCTCGCGCCATCAGACGAGCTTAGGCGTATCACGAGGCCCTTTCGTCTTC
full-500 B5	aJK4.L.35	GCCTCCCTCGCGCCATCAGATCACGTTAGGCGTATCACGAGGCCCTTTCGTCTTC
full-150 B1	aJK4.L.36	GCCTCCCTCGCGCCATCAGTCAGTGTTAGGCGTATCACGAGGCCCTTTCGTCTTC
full-150 B2	aJK4.L.37	GCCTCCCTCGCGCCATCAGCACGTGCTAGGCGTATCACGAGGCCCTTTCGTCTTC
full-150 B3	aJK4.L.38	GCCTCCCTCGCGCCATCAGTCTCATATAGGCGTATCACGAGGCCCTTTCGTCTTC
full-150 B4	aJK4.L.39	GCCTCCCTCGCGCCATCAGCTCGAGTTAGGCGTATCACGAGGCCCTTTCGTCTTC
full-150 B5	aJK4.L.40	GCCTCCCTCGCGCCATCAGTGACGACTAGGCGTATCACGAGGCCCTTTCGTCTTC
full-0 B1	aJK4.L.41	GCCTCCCTCGCGCCATCAGAGCATACTAGGCGTATCACGAGGCCCTTTCGTCTTC
full-0 B2	aJK4.L.42	GCCTCCCTCGCGCCATCAGTCGTAGTTAGGCGTATCACGAGGCCCTTTCGTCTTC
full-0 B3	aJK4.L.43	GCCTCCCTCGCGCCATCAGACTCTCATAGGCGTATCACGAGGCCCTTTCGTCTTC
full-0 B4	aJK4.L.44	GCCTCCCTCGCGCCATCAGCTCACTCTAGGCGTATCACGAGGCCCTTTCGTCTTC
full-0 B5	aJK4.L.45	GCCTCCCTCGCGCCATCAGACGACTGTAGGCGTATCACGAGGCCCTTTCGTCTTC
control 1	aJK4.L.46	GCCTCCCTCGCGCCATCAGACTCGATTAGGCGTATCACGAGGCCCTTTCGTCTTC
control 2	aJK4.L.47	GCCTCCCTCGCGCCATCAGCTACGTCTAGGCGTATCACGAGGCCCTTTCGTCTTC

1d. Processing of sequence reads

We used a variety of quality filters for processing raw sequence reads into a list of mutant promoter sequences σ and corresponding FACS batches μ. Of the 448,416 reads produced by 454 pyrosequencing, 435,988 contained the 10-mer AGCGCAACGC, which is located at positions [-85:-76] of the wild-type *lac* promoter. 362,646 reads also contained the 10-mer AATTGTGAGC, which is located at region [+1:+10]. These two 10-mers were separated by 75 bp in 314,915 sequences. This 75 bp region contained no Ns in 312,229 sequences. Finally, 308,309 sequences satisfied all of the above conditions, and further contained a valid barcode. Sequence reads that satisfied these requirements had length 242.3 ± 1.6 bp; this variation in read length is expected as a natural consequence of how 454 pyrosequencing works. Section 1f describes how this list of 312,229 sequences was further pruned to remove redundant sequence reads, thereby yielding six data sets comprising 220,591 sequences σ and corresponding measurements μ.

1e. Library substitution rates

The observed substitution rates for the full-wt, crp-wt, and rnap-wt libraries, shown in Table S3, closely match the design specifications of each library. All rates are computed from the sequences in each experiment's B0 batch — the batch of cells randomly sampled from each library. Statistics for both the intentionally mutated regions and the regions that were designed to remain wild-type are shown. For each region we list the intended substitution rate, the substitution rate observed in sequence reads (mean ± RMSD over all indicated positions), as well as the substitution rate observed at positions within the indicated region that have a specific wild-type base.

Table S3: Observed substitution rates

Batch	Region	Target sub. %	Observed sub. %	Observed sub. % by wild-type base			
				A	C	G	T
full-wt B0	[-75:-1]	12	11.6 ± 1.9	12.0 ± 0.5	13.8 ± 0.8	12.6 ± 0.5	9.2 ± 0.4
crp-wt B0	[-74:-49]	24	23.3 ± 3.4	25.2 ± 1.0	27.3 ± 0.4	24.9 ± 0.5	19.4 ± 1.0
crp-wt B0	[-75],[-48:-1]	0	0.28 ± 0.17	0.28 ± 0.13	0.47 ± 0.16	0.23 ± 0.12	0.16 ± 0.08
rnap-wt B0	[-39:-4]	15	15.2 ± 2.0	16.1 ± 0.7	16.2 ± 0.6	17.4 ± 1.1	13.1 ± 0.5
rnap-wt B0	[-75:-40],[-3:-1]	0	0.17 ± 0.08	0.20 ± 0.06	0.22 ± 0.05	0.13 ± 0.05	0.11 ± 0.10

The variation among positions that share a given wild-type base is roughly consistent with the expected statistical fluctuations (~0.5%) due to each B0 batch containing only ~5,000 sequences. However, the substitution rates for positions corresponding to different wild-type bases show significant systematic differences. This position-to-position variation is probably due primarily to variations in the nucleoside phosphoramidite mixtures used for synthesizing each position.

Importantly, however, positions at which no variation was specified in the design of the sequence libraries still showed a nonzero substitution rate, ranging between 0.1% and 0.5%. By contrast, the observed substitution rate within the control amplicon libraries, which were prepared directly from pUA66-lacZ plasmid (containing the wild-type *lac* promoter), was 0.017 ± 0.026 %. This difference suggests that the substitutions observed in the supposedly wild-type regions of crp-wt and rnap-wt did not result from sequencing errors or errors in amplicon PCR, but rather from substitutions that occurred either during DNA synthesis or during the construction of reporter construct libraries. Supporting evidence for this hypothesis — that these undesired substitutions occurred prior to FACS — is given by Fig. S3B, which shows that observed substitutions at positions -37 and -14 in the crp-wt experiment are significantly informative about flow cytometry measurements.

1f. Post-sort loss of library diversity

The validity of our analysis requires that all entries in our data sets arise from different FACS-sorted cells and thereby represent independent measurements of promoter activity. It was therefore important that we not only FACS-sort a sufficient number of cells, but that enough of these cells actually produce sequenced amplicons.

In our six experiments we sorted 100,000 flow-cytometer-detected "events" into each FACS batch. Plating 0.1% of the cells in each batch immediately afterward, we observed ~70 equal-sized colonies per plate. We concluded that each batch received ~70,000 viable cells — i.e. that ~70% of detected "events" represented viable cells. It appears, however, that a large loss in library diversity occurred between the sorting of these cells and the sequencing of amplicons, and that as a result our experiment yielded amplicons from only ~13,000 independently sorted cells per batch. We determined this library reduction as follows.

After FACS, we miniprepped the plasmids within each batch, used these plasmids as template for amplicon PCR, then sequenced the resulting amplicons. If a particular batch originally contained N independently sorted (viable) cells, each cell containing a different CRM sequence, and n sequences in our final data set derived from this batch, the probability p_k of a given sequence occurring k times in this batch should follow what one would expect from sampling n out of N objects with replacement. Using a Poisson approximation to this binomial probability (a valid approximation because $N, n >> 1$),

(S1)
$$p_k = \binom{n}{k}\left(\frac{1}{N}\right)^k\left(1-\frac{1}{N}\right)^{n-k} \approx \left(\frac{n}{N}\right)^k \frac{e^{-n/N}}{k!}.$$

Now consider the probability f_k of a sequence randomly drawn from this batch having exactly k occurrences therein:

(S2)
$$f_k = \frac{kp_k}{\sum_{k'=1}^{n} k'p_{k'}} \approx \frac{kp_k}{\sum_{k'=1}^{\infty} k'p_{k'}} = \frac{Nk}{n} p_k .$$

In particular, consider

(S3)
$$f_1 = e^{-n/N} \quad \text{and} \quad f_2 = \frac{n}{N} e^{-n/N}.$$

These quantities, f_1 and f_2, are readily determined from our data sets. Since we also know n, we can use f_1 and f_2 to estimate the effective population size N^{eff} of sorted cells that gave rise to sequenced amplicons. In fact, this can be done in multiple ways:

(S4)
$$N^{eff,1} = -\frac{n}{\log(f_1)} \quad \text{and} \quad N^{eff,2} = n\frac{f_1}{f_2}.$$

For the full-wt, crp-wt, and rnap-wt experiments, Table S4 shows the total number (n) of sequences found in batch B0, the number of distinct sequences in this batch, the quantities f_1 and f_2, the two resulting estimates $N^{eff,1}$ and N^{eff}, and the total number of distinct sequences observed across all batches B0-B9.

Table S4: Post-sort library reduction

Data set	Total in B0	Distinct in B0	f_1	f_2	$N^{eff,1}$	$N^{eff,2}$	Distinct in B0-B9
full-wt	7,680	5,799	0.57	0.26	13,946	16,801	50,518
crp-wt	5,805	4,347	0.56	0.26	10,256	12,328	45,050
rnap-wt	6,453	4,748	0.55	0.25	10,926	13,998	43,818

In all three cases, both $N^{eff,1}$ and $N^{eff,2}$ are ~12,000. By contrast, each experiment generated ~45,000 distinct sequences across all 10 batches — nearly 10 times the

number of distinct sequences within each individual batch. We therefore concluded that, prior to sorting, the library size in each experiment was >> 45,000, whereas after sorting the number of distinct sequences within each batch dropped from ~70,000 to ~12,000. Similar N^{eff} values were found for each of the 45 batches in our six experiments, suggesting uniform library diversity loss across all FACS batches.

Because of this reduction in library diversity, a single sequence appearing multiple times in the same batch was most likely to result from a single sorted cell giving rise to multiple sequenced amplicons, not from different sorted cells each giving rise to one sequenced amplicon. In other words, the appearance of one sequence multiple times in a single batch would represent a single measurement, not multiple independent measurements. To ensure that our analysis used only independent measurements, we performed computations using data sets in which each distinct sequence in each batch was listed only once. This left a total of 220,591 non-redundant sequences across our six experiments; the number of such sequences remaining in each experiment's data set is shown in Table 1 of the main text.

2. Supporting theoretical methods

2a. Overview of mutual information

Mutual information provides a fundamental measure of dependence between any two random variables, say x and y. The mutual information I(x;y) between two such variables is computed from their joint distribution $p(x,y)$ as

(S5) $$I(x;y) = \sum_x \sum_y p(x,y) \log_2 \frac{p(x,y)}{p(x)p(y)} .$$

If either variable is continuous, the corresponding summation is replaced by an integral. $I(x;y)$ can thus be computed regardless of whether both variables are discrete (such as a base b and batch μ), one variable is discrete and the other is continuous (such as predicted binding energy ε and batch μ), or both variables are continuous. $I(x;y)$ is clearly symmetric, i.e. $I(x;y) = I(y;x)$. Also, a basic result of information theory is that $I(x;y) \geq 0$, with zero information only if x and y are independent, i.e. $p(x,y) = p(x)p(y)$.

Another basic property of mutual information is that $I(x;y)$ doesn't change if either x or y undergoes an invertible transformation. To see this, consider some invertible transformation $z(x)$ of x. Then I(z;y) = I(x;y) (regardless of whether y is continuous or discrete) because

$$
\begin{aligned}
I(z;y) &= \int dy \int dz\, p(z,y) \log_2 \frac{p(z,y)}{p(y)p(y)} \\
&= \int dy \int dx \left|\frac{dz}{dx}\right| p(x,y) \left|\frac{dz}{dx}\right|^{-1} \log_2 \frac{p(x,y)\left|dz/dx\right|^{-1}}{p(x)p(y)\left|dz/dx\right|^{-1}} \\
&= \int dy \int dx\, p(x,y) \log_2 \frac{p(x,y)}{p(x)p(y)} \\
&= I(x;y).
\end{aligned}
\tag{S6}
$$

This has important implications for what kinds of parameters we can infer when fitting models to our data (see Sec. 2c).

Mutual information provides a measure of dependence, much as correlation does. In fact, if x and y are two correlated Gaussian random variables, there exists a simple relationship between $I(x;y)$ and the correlation coefficient ρ between x and y. To see this, assume that both x and y have mean 0 and variance 1. Then

$$
p(x,y) = \frac{1}{2\pi\sqrt{|\Lambda|}} \exp\left(-\frac{1}{2} \begin{bmatrix} x & y \end{bmatrix} \Lambda^{-1} \begin{bmatrix} x \\ y \end{bmatrix} \right) \quad \textit{where} \quad \Lambda = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}. \tag{S7}
$$

Plugging this into Eq. S5 gives

$$
I(x;y) = -\frac{1}{2} \log_2 (1-\rho^2). \tag{S8}
$$

Note that $I(x;y) = 0$ for $\rho = 0$, and $I(x;y) \to \infty$ as $\rho \to \pm 1$. Because $I(x;y)$ is invariant under any affine transformation of x or y (via Eq. S6), Eq. S8 holds for all correlated Gaussian variables x and y with correlation coefficient ρ.

Despite the simplicity of Eq. S5, actually estimating mutual information from finite data is nontrivial. The steps we took to do this are described in Sec. 2e.

2b. Statistical inference using mutual information (justification of Eq. 1)

Our rationale for sampling model parameters according to Eq. 1 comes from a relationship between mutual information and a quantity called "error-model-averaged likelihood", first described in Kinney et al., 2007. Kinney et al. considered the problem of fitting a sequence-dependent model of some biological activity x to noisy measurements μ of this activity for a large number N of sequences σ. They considered a particular case where the sequences σ were the intergenic regions of *S. cerevisiae* and the measurements μ were the fluorescence intensities observed in ChIP-chip (Lee et al., 2002) or protein binding microarray (Mukherjee et al., 2004) experiments on yeast

transcription factors. The models they considered consisted of energy matrices together with energy thresholds. Model predictions were binary — "bound" or "not bound" — corresponding to whether an intergenic sequence contained at least one site with energy below the specified threshold. Despite the specific nature of their interests, though, the inference issues Kinney et al. addressed were quite general.

To determine model parameters θ, one can sample θ according to the likelihood of actually seeing the set of observations $\{\mu_i\}$ given the model predictions $\{x_i\}$, where the index i=1,...,N runs over all the sequences assayed, and x_i is the prediction assigned to sequence σ_i. Normally, one would compute this likelihood using prior knowledge of an "error model" $p(\mu|x)$, which specifies the distribution of measurements μ to be expected given a particular value for the underlying quantity-of-interest x. Likelihood is expressed in terms of this error model according to the standard formula,

$$p(\{\mu_i\} \mid \{\sigma_i\},\theta) = p(\{\mu_i\} \mid \{x_i\}) = \prod_{i=1}^{N} p(\mu_i \mid x_i)\,. \tag{S9}$$

For Kinney et al., $p(\mu|x)$ represented the distribution of microarray intensities expected from sequences that either contained (x = "bound") or did not contain (x = "not bound") a protein binding site.

Since Kinney et al. had no *a priori* knowledge of the experimental error model $p(\mu|x)$, they instead proposed the use of "error-model-averaged" likelihood, computed by averaging the standard likelihood (Eq. S9) over all possible error models $p(\mu|x)$ according to an explicit prior on the space of possible error models. They showed that, under very general assumptions, error-model-averaged likelihood has a remarkably simple structure:

$$p_{ema}(\{\mu_i\} \mid \{\sigma_i\},\theta) = \left\langle \prod_{i=1}^{N} p(\mu_i \mid x_i) \right\rangle_{\text{all possible } p(\mu|x)} = 2^{N[I(x;\mu)-H(\mu)-\Delta]}\,. \tag{S10}$$

where $I(x;\mu)$ is the mutual information between measurements μ and model predictions x, $H(\mu)$ is the entropy of the experimental measurements (which does not depend on model parameters θ) and Δ is a positive number that depends on the particular prior used to compute the average $\langle\cdots\rangle$ over possible error models $p(\mu|x)$. Kinney et al. further showed that, under mild assumptions on the prior, $\Delta \to 0$ as $N \to \infty$, meaning that Δ can be ignored in the large N limit.

Finally, Kinney et al. showed that EMA likelihood could be computed exactly if one assumed a Dirichlet prior on the space of error models. However, the resulting formula for EMA likelihood caused problems in the analysis presented in the present paper; it gave likelihood landscapes that were unreasonably rugged and thus prevented us from

optimizing model parameters. This ruggedness probably stems from the fact that the Dirichlet prior used by Kinney et al. does not require that $p(\mu|x)$ be similar for similar values of x (and fixed μ). By contrast, any biophysically reasonable prior on the space of error models should require such smoothness in the *x*-direction.

We therefore took a different approach that, while not proven mathematically, works very well in practice: we assumed that error-model-averaged likelihood could be safely approximated as

(S11) $$p_{ema}(\{\mu_i\} \mid \{\sigma_i\}, \theta) = \text{const} \times 2^{NI_{smooth}(x;\mu)}.$$

where $I_{smooth}(x;\mu)$ is computed using a Gaussian-smoothed version of the joint distribution $p(\mu,x)$ in the formula for mutual information (Eq. S5). Eq. S11 is just a more explicit form of Eq. 1 in the main text. Sec. 2e describes how this smoothing of $p(\mu,x)$ was done.

2c. Maximizing mutual information leaves some model parameters undetermined

Our inability to determine certain model parameters — the multiplicative scale of ε_c and ε_r when fitting individual energy matrices, and τ_{max} and C_r when fitting τ — arises from a basic property of mutual information: transforming model predictions *x* in a way that does not change their rank order leaves the predictive information $I(x;\mu)$ unchanged (see Eq. S6). Thus multiplying ε_c or ε_r by a constant leaves $I(\varepsilon_c;\mu)$ and $I(\varepsilon_r;\mu)$ invariant. Similarly, changing τ_{max} has no effect on $I(\tau;\mu)$.

It is also true that changing C_r has no effect $I(\tau;\mu)$, though this is not as obvious. Eq. 2 of the main text can be rewritten

(S12) $$\tau = \frac{\tau_{\max}}{1 + C_r \eta^{-1}} \quad \text{where} \quad \eta = e^{-\varepsilon_r/RT} \frac{1 + C_c e^{-(\varepsilon_c + \varepsilon_l)/RT}}{1 + C_c e^{-\varepsilon_c/RT}}.$$

Here, τ is an invertible function of η for all $C_r > 0$, and so $I(\tau;\mu) = I(\eta;\mu)$ by Eq. S6. But the expression for η makes no reference to C_r. The value of C_r therefore has no influence on $I(\tau;\mu)$, and cannot be inferred by fitting τ to data.

It may seem odd that inferring models for ε_c and ε_r by fitting τ allowed us to determine these energies in kcal/mol, whereas separately fitting models for these binding energies did not. In general, which parameters one can or cannot determine depends on the specific analytic form of the model being considered. We have found that simpler (more linear) models tend to have more unconstrained parameters. We cannot yet make definitive statements on this matter, though, and which parameters one can or cannot fit in any given model must be assessed on a case-by-case basis. Kinney, 2008, shows how to perform this assessment analytically.

2d. Fitting a model to multiple data sets

A single biophysical model can be fit to multiple data sets using a straightforward generalization of Eq. 1. One reason for doing so is that, by using all the data in hand, one can minimize error bars on inferred model parameters. More compelling, though, is the fact that letting select parameters vary between data sets can shed light on the molecular mechanisms responsible for modulating transcriptional activity in response to different inducing conditions or genomic backgrounds.

Generalizing the relationship between mutual information and likelihood to data from multiple experiments is straightforward:

(S13) $$p(\text{multiple data sets}|\text{model}) = \prod_j p(\text{data set j}|\text{model}) = \text{const} \times 2^{\sum_j N_j I(x_j;\mu_j)}$$

where j indexes different data sets, x_j denotes model predictions specific to experiment j, μ_j denotes corresponding measurements, and N_j is the number of independent measurements in that data set. This relation is valid regardless of which parameters are experiment-specific and which are shared across all experiments. Model parameters can be sampled according to this distribution the same way they are sampled according to Eq. 1.

2e. Computing mutual information

Despite the fundamental role that mutual information plays in information theory, there are a variety of ways to estimate mutual information from finite amounts of data. Which technique one should use depends on the specific context of the calculation. In this paper, there are three contexts in which we compute mutual information, each requiring its own method:

i. Computing $I(b;\mu)$ for information footprints
ii. Comparing $I(\tau;\mu)$ *to* $I(\varepsilon_r,\varepsilon_c;\mu)$
iii. Computing $I(x;\mu)$ for sampling model parameters

<u>*Computing I(b;μ)* for *information footprints*</u>

We used the following procedure (schematized in Fig. S6A) to compute the $I(b;\mu)$ values in information footprints. For a given nucleotide position within the aligned N sequences σ, we determined the fraction of the time that base b occurred at that position in a sequence recovered from batch μ. From this occurrence frequency, $f(b,\mu)$, one can compute the "naive" mutual information between bases b and batches μ:

(S14) $$I_{naive}(b;\mu) = \sum_{b,\mu} f(b,\mu) \log_2 \frac{f(b,\mu)}{f(b)f(\mu)},$$

where $f(b)$ and $f(\mu)$ are the marginal distributions of $f(b,\mu)$. However, finite sample effects will typically cause Eq. S14 to overestimate the true mutual information — i.e. the value of $I(b;\mu)$ one would find if one was able to collect an infinite amount data. If $f(b,\mu)$ is well-sampled at all values of b and μ, though, one can analytically compute the correction to this information value (Treves and Panzeri, 1995),

$$\text{(S15)} \qquad I(b;\mu) = \sum_{b,\mu} f(b,\mu)\log_2 \frac{f(b,\mu)}{f(b)f(\mu)} - \frac{(n_b-1)(n_\mu-1)\log_2 e}{2N} + o(N^{-2}),$$

where $n_b = 4$ is the number of possible bases b and n_μ is the number of batches μ (5 or 10 in our experiments). Using the first two terms on the right-hand-side of Eq. S15 thus allows one to compute an estimate of $I(b;\mu)$ that is essentially unbiased. This is the formula we used to estimate all $I(b;\mu)$ values shown in Fig. 2 and Fig. S3.

To determine error bars on each of these $I(b;\mu)$ information values, we computed a large number of naive information estimates $I^{50\%}_{naive}(b;\mu)$ evaluated on random subsamples of the data consisting of half the sequences σ. The variance in these values across different subsamples was used to compute the uncertainty in $I(b;\mu)$:

$$\text{(S16)} \qquad \delta I(b;\mu) = \frac{1}{\sqrt{2}}\sqrt{\mathrm{var}\left(I^{50\%}_{naive}(b;\mu)\right)}.$$

<u>*Comparing $I(\tau;\mu)$ to $I(\varepsilon_r,\varepsilon_c;\mu)$*</u>

As in our computation of $I(b;\mu)$, we looked for absolute information values with error bars when comparing $I(\tau;\mu)$ *to* $I(\varepsilon_r,\varepsilon_c;\mu)$. There were two new issues we had to deal with, however.

The first is that model predictions x are continuous values, not categorical variables (like bases b), and some regularization method was required to estimate a well-sampled joint distribution $p(x,\mu)$. To compute $I(\tau;\mu)$, we took advantage of the fact that transformations of τ that do not change the rank order of these predictions do not affect $I(\tau;\mu)$. Specifically, we equipartitioned τ values among 100 bins, thus reassigning each prediction x to a bin number $X=1,2,...,100$ according to its rank order. To compute $I(\varepsilon_c,\varepsilon_r;\mu)$, we equipartitioned both ε_c and ε_r into 15 bins each, then used X to number each of the resulting 225 bins. The observed frequency distribution $f(X,\mu)$ between these bin numbers and batches μ was then used to compute $I_{naive}(X;\mu)$ in both cases.

The second issue is that, unlike in estimating $I(b;\mu)$, the frequency table $f(X,\mu)$ typically did not contain many counts for every pair of X and μ. The analytic correction in Eq. S15, used for countering the upward bias in the naive estimate of mutual information, was therefore invalid. Instead we used the "direct" method of information estimation described by Slonim et al., 2005 (following the initial proposal by Strong et al., 1998). We computed $f(X;\mu)$ and the corresponding value for $I_{naive}(X;\mu)$ for 25 subsamples of

the data for each of the three subsample percentages: 83%, 63%, and 50%. The interpolation scheme described in Slonim et al. was then used to read off *I(X;μ)* in the infinite data limit from the mean values of $I_{naive}^{83\%}(X;\mu)$, $I_{naive}^{63\%}(X;\mu)$, and $I_{naive}^{50\%}(X;\mu)$. Error bars on *I(X;μ)* were computed in the same way as for information footprints:

(S17) $$\delta I(X;\mu) = \frac{1}{\sqrt{2}}\sqrt{\mathrm{var}\left(I_{naive}^{50\%}(X;\mu)\right)}.$$

<u>*Computing I(x;μ) for sampling model parameters*</u>

The method of Slonim et al., 2005, is too computationally demanding to use as part of the Monte Carlo procedure we used to infer model parameters. We therefore used an alternate procedure (Fig. S6B) to determine *I(x;μ)* for use in Eq. 1. After assigning a rank order R to each model prediction we convolved the observed frequency distribution *f(R,μ)* in the *R* dimension with a Gaussian having a standard deviation equal to 4% the number of data points. This generated a regularized distribution $p(R,\mu)$, which we used in Eq. S5 to compute *I(R;μ) = I(x;μ)*.

2f. Parallel tempering Monte Carlo sampling of model parameters

We used a custom parallel tempering Monte Carlo algorithm (Earl and Deem, 2005) to sample parameters according to their likelihood in Eq. S11. We note that it is valid to assume $p(\theta|\{\mu_i\},\{\sigma_i\}) = const \times p(\{\mu_i\}|\{\sigma_i\},\theta)$ if there is no need for a strong prior distribution $p(\theta)$ on model parameters θ. The algorithm that follows can be readily altered to accommodate such a prior, however, if one is needed.

The parallel tempering Monte Carlo algorithm we used to sample parameters θ worked as follows.

1. We initiated *K* processes (using *K* = 40 or 100). For each process *k* = 0,1,...,*K*-1 we
    a. Chose an initial value for model parameters θ_k
    b. Computed the mutual information $I(x_k;\mu)$ resulting from this choice of parameters θ_k
    c. Assigned an "inverse temperature" $\beta_k \leq 1$ to each process *k*, with $\beta_0 = 1$ for process *k* = 0. We chose $\log_{10} \beta_k$ values to be evenly spaced between 1 and either 3 or 4

2. We then iterated the following steps simultaneously for all processes *k*.
    a. Randomly perturbed model parameters $\theta_k \to \theta_k'$
    b. Computed the predictive information $I(x_k';\mu)$ resulting from perturbed parameters θ_k'
    c. Replaced θ_k with θ_k' with probability $\min(1, 2^{\beta_k N[I(x_k';\mu)-I(x_k;\mu)]})$

d. Recorded θ_k

3. During this iteration, we would periodically choose two processes, $k,m < K$ and switch parameter values $\theta_k \leftrightarrow \theta_m$ with probability $\min(1, 2^{-N[\beta_k - \beta_m][I(x_k;\mu) - I(x_m;\mu)]})$. This is the defining step in parallel tempering Monte Carlo algorithm (Earl and Deem, 2005).

4. We used the resulting ensemble of parameter values θ_0 from process $k = 0$ to determine the best value, the mean value, and the RMSD uncertainty in each parameter in θ. Parameter ensembles from all other processes $k = 1,2,...,K$-1 were used only for diagnostic purposes, not for estimating model parameters.

In other words, this algorithm uses multiple particles (K of them) to explore the probability landscape (over the space of model parameters θ) defined by Eq. 1. Different particles have different "temperatures"; hotter particles (lower β_k) experience a smoother landscape and are less prone to get stuck in local optima, while cooler particles (higher β_k) are more sensitive to the shape of the landscape and can locate optima more precisely that hotter particles. Most of the time, particles explore the probability landscape according to the normal Metropolis algorithm (step 2). The power of parallel tempering comes from the periodic execution of step 3. This allows hotter particles and cooler particles to trade parameter values if the hotter particle is found to be in a higher probability state, since such a situation would suggest that the cooler particle was stuck in a local optimum and that the hotter particle had found a better solution. Parallel tempering Monte Carlo thereby provides rigorous sampling of parameters θ while enabling the quick identification of global optima even in the presence of many local optima; normal Monte Carlo, on the other hand, is very prone to getting stuck in local optima.

2g. Model comparison to literature

Performance of CRP and RNAP energy matrices on our six data sets

Fig. S7 compares the performance of our inferred CRP and RNAP energy matrices to three CRP matrices and three RNAP matrices found in the literature. Every one of our inferred matrices performs better on every one of our data sets (with two exceptions) than any of the matrices taken from the literature. One exception is that the CRP energy matrix fit to full-wt data performs equally well on full-0 data as the model of Gunasekera et al.,1992.

The other exception is the performance of CRP matrices on the rnap-wt data set. The information values in this column are somewhat suspect, though, because the CRP binding site was not intentionally mutagenized in the rnap-wt experiment; listed information values (all less than 0.001 bits) are significantly nonzero only because a small fraction of sequences from the rnap-wt experiment have spurious mutations in their CRP binding sites (see Table S3). The dearth of sequences with mutated CRP

binding sites may have caused problems for our estimation of $I(\varepsilon_c;\mu)$. This same concern applies to the performance of RNAP matrices on crp-wt data.

CRP and RNAP matrices gleaned from the literature

The energy matrices derived from Harley and Reynolds, 1987, Lisser and Margalit, 1993, and Robison et al., 1998 were constructed from either aligned CRP binding sites or from aligned σ^{70} promoters (having an 18 bp spacing between their -10 and -35 elements) using the method of Berg and von Hippel (Berg and von Hippel, 1987; Stormo, 2000): for each list of aligned sequences, energy matrix elements E_{bi} were computed (in arbitrary units) as

(S18) $$E_{bi} = -\log\frac{f_i(b)}{p(b)}, \quad f_i(b) = \frac{c_i(b)+1}{N+4},$$

where $c_i(b)$ is the number of sites that have base b at position i, N is the total number of sites, and $p(b)$ is the frequency of base b in the intergenic DNA of *E. coli* (which has a GC content of 43.4%).

The RNAP energy matrix reported in Kiryu et al. (2005) was derived from microarray expression data using an algorithm written by the authors; matrix elements were gleaned from the bottom panel of Fig. 4 in that publication.

The CRP energy matrix reported in Nagaraj et al. (2008) was found using the QPMEME algorithm (Djordjevic et al., 2003) applied to CRP SELEX data.

The CRP energy matrix quoted for Gunasekera et al. (1992) was constructed from the measured ΔΔG values listed in Table 1 of that publication; ΔΔG = 5.2 kcal/mol was assumed for values listed as ΔΔG > 5.2 kcal/mol.

Thermodynamic parameters from Kuhlman et al., 2007

The value of f_{cAMP} = 240 ± 13, listed for TK310 cells in Table 2 of Kuhlman et al., 2007, corresponds to $\exp[-\varepsilon_i / RT]$ with a CRP-RNAP interaction energy of ε_i = -3.37 ± 0.03 kcal/mol. Kuhlman et al. also predicts that the concentration of active intracellular CRP in TK310 cells is related to the concentration of cAMP used to supplement growth media via C_c = [cAMP]/(320 μM). This implies that $C_c^{full\text{-}500} = 10^{0.19}$ and $C_c^{full\text{-}150} = 10^{-0.33}$, both of which are about 20-times larger than our inferred values (Fig. 4B, main text).

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4. Supporting information figures

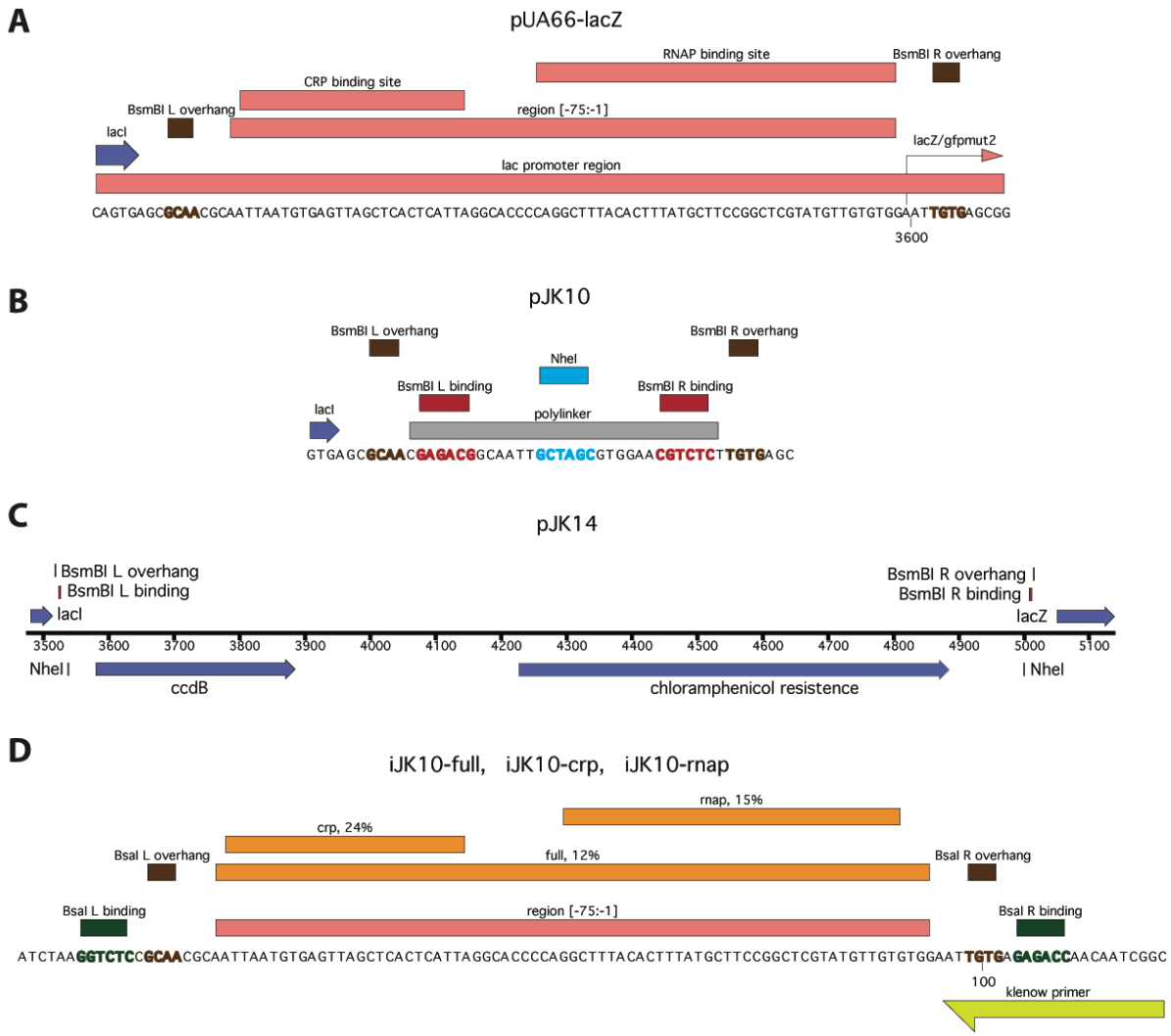

Figure S1. Construction of plasmid library. A) *Lac* promoter region of the plasmid pUA66-lacZ. B) Homologous region of pJK10, a plasmid derived from pUA66-lacZ in which region [-75:-1] of the *lac* promoter was replaced with a polylinker sequence containing outward-facing BsmBI sites and a central NheI site. C) Corresponding region of the vector plasmid pJK14, which was derived from pJK10 by insertion at the NheI site of a DNA cassette containing the *ccdB* and chloramphenicol resistance genes. D) Insert libraries iJK10-full, iJK10-crp, and iJK10-rnap. The mutagenized region of these inserts is indicated in orange. In all panels, bold red sequences indicate BsmBI binding sites, bold green sequences indicate BsaI binding sites, and bold brown sequences indicate the 4 bp 5' overhang at which BsaI-digested iJK10 inserts were inserted into BsmBI-digested pJK14 vector. The plasmid library created using iJK10-full inserts was used in the full-wt, full-500, full-150, and full-0 experiments. Plasmid libraries used in the crp-wt and rnap-wt experiments were created in the same manner using iJK10-crp or iJK10-rnap, which had different mutagenized regions (panel D).

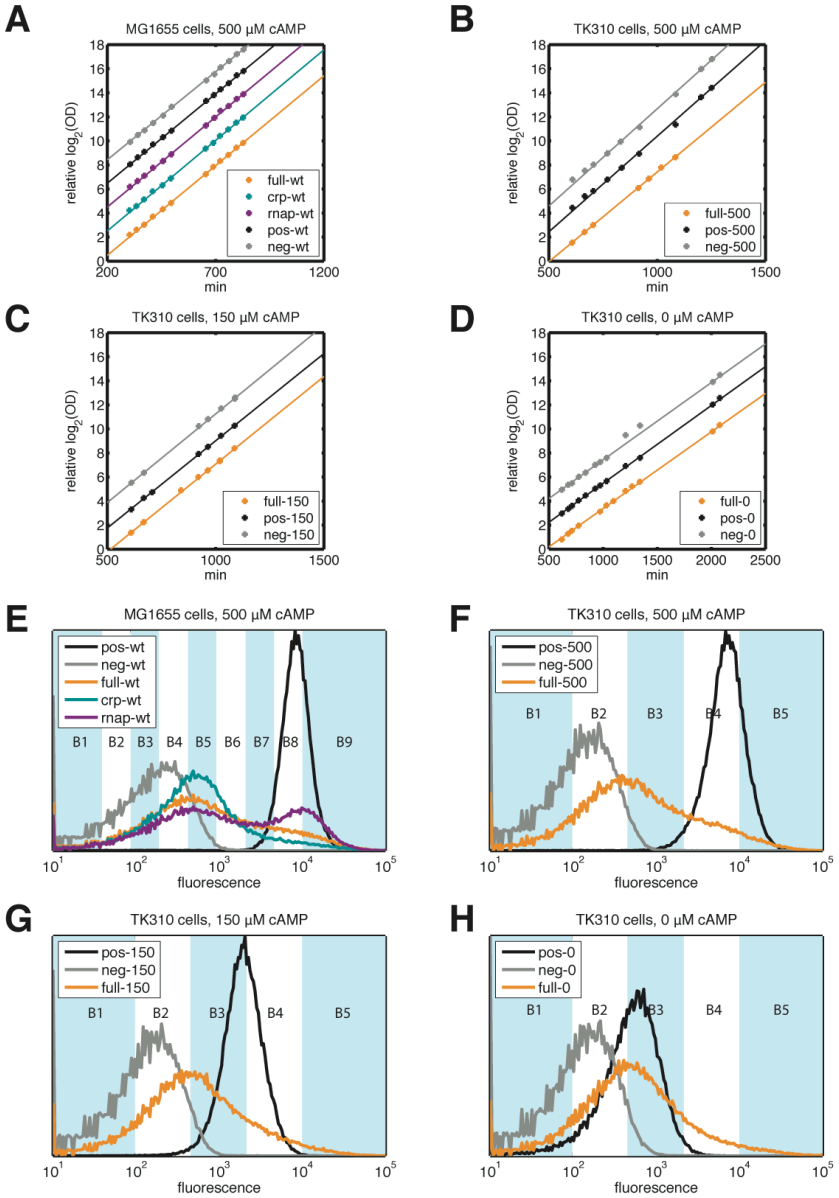

Figure S2. Library induction and sorting. A) Growth curves of full-wt, crp-wt, and rnap-wt cell libraries during induction of *gfpmut2* expression. Also shown are growth curves for pos-wt and neg-wt cells — MG1655 cells carrying the positive control (pUA66-lacZ) and negative control (pJK10) plasmids respectively. B,C,D) Growth curves for the full-500, full-150, and full-0 libraries, as well as TK310 cells carrying positive or negative control plasmids. Panels B, C, and D show growth in 500 μM, 150 μM, and 0 μM cAMP, respectively. E) Post-induction FACSVantage fluorescence distributions of full-wt, crp-wt, and rnap-wt cell libraries, as well as pos-wt and neg-wt controls. Each library was sorted into 10 batches: batch B0 received randomly selected cells, while batches B1-B9 received cells from the indicated fluorescence ranges. F) Fluorescence distribution of the full-500 library (TK310 cells induced with 500 μM cAMP), as well as of the corresponding positive and negative control populations. full-500 cells were sorted into the five batches, B1-B5, as indicated. G,H) Analagous distributions and fluorescence gates for the full-150 library (induced with 150 μM cAMP) and full-0 library (induced with 0 μM cAMP). As explained in the main text, 1 mM IPTG was used for the induction of all cell libraries and control populations.

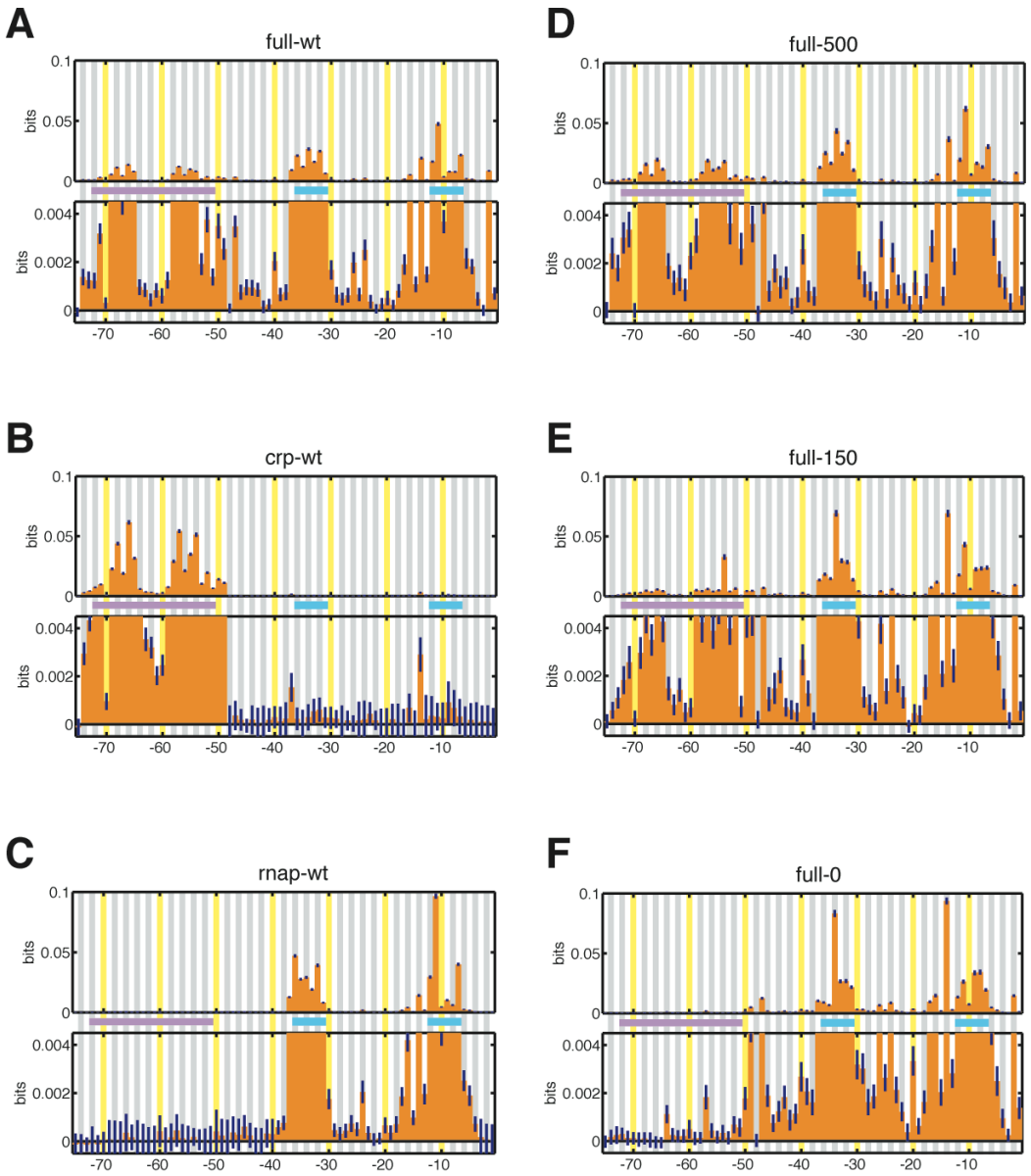

Figure S3. Information footprints from all six experiments: A) full-wt, B) crp-wt, C) rnap-wt, D) full-500, E) full-150, and F) full-0. As in the main text, the lower plot in each panel is a 20X magnification of the upper plot, and dark blue lines indicate uncertainties due to finite sample effects. The CRP binding site and -10 and -35 elements are also indicated as in the main text. We call attention to panels D, E, and F, which show that the concentration of cAMP (500 μM, 150 μM, and 0 μM respectively) used to induce reporter expression in TK310 cells correlates with the magnitude of $I(b;\mu)$ observed within CRP's binding site (positions [-72:-51]).

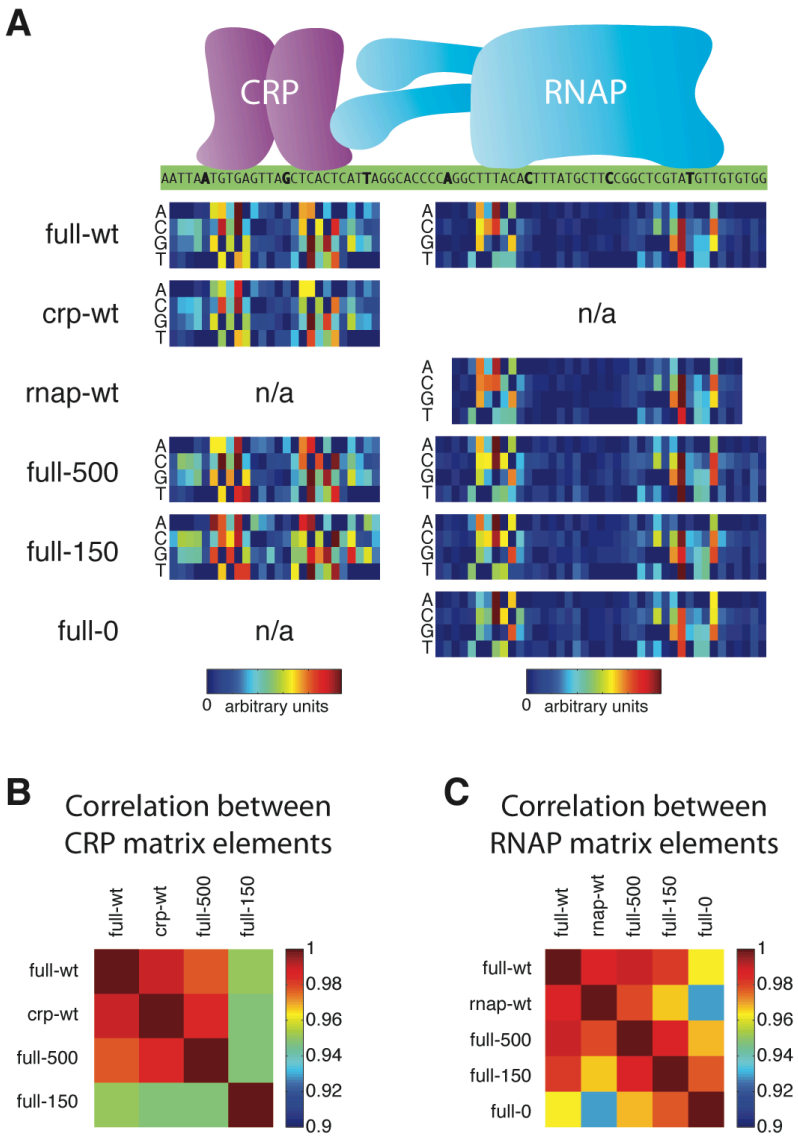

Figure S4. Inferred energy matrices. A) CRP and RNAP energy matrices inferred by maximizing $I(\varepsilon_c;\mu)$ or $I(\varepsilon_r;\mu)$ on each of our six data sets. CRP energy matrices were fit to the 26 bp site [-74:-49], which encompasses 2 bp on either side of the 22 bp CRP site [-72:-51] typically cited in the literature. No RNAP energy matrix could be fit to positions [-41:-1] using crp-wt data, since only positions [-74:-49] were intentionally mutagenized in this experiment (though mutations were detected at other positions; see Table S3). Similarly, the fact that only positions [-39:-4] were specifically mutated in the rnap-wt experiment prevented us from fitting a CRP energy matrix and reduced the extent of the RNAP matrix we were able to infer. In the full-0 experiment, there was not enough active CRP (owing to a lack of cAMP in the induction media) for us to infer a CRP energy matrix. B) Correlation coefficients between elements of the optimal CRP energy matrices inferred from full-wt, crp-wt, full-500, and full-150 data; all of these correlations are > 0.95, indicating great concordance between the CRP matrices inferred from different experiments. C) Correlation coefficients between the elements of optimal RNAP energy matrices inferred from the full-wt, rnap-wt, full-500, full-150, and full-0 experiments. All of these correlations are > 0.92, and most are > 0.97, again signaling excellent agreement between experiments.

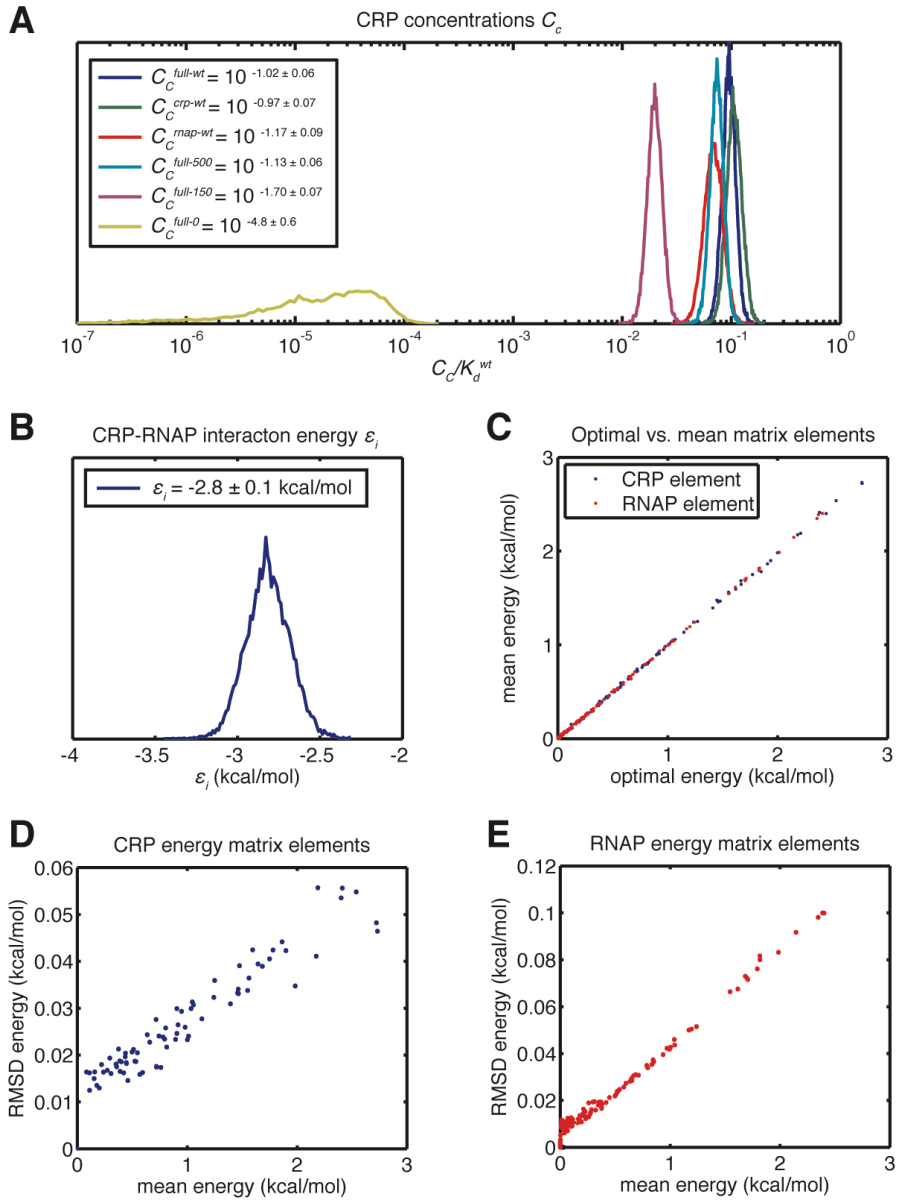

Figure S5. Final model parameter distributions. A-E) Monte Carlo distributions sampled for the parameters of our final model for τ, which was fit to all six data sets using one value for the CRP-RNAP interaction energy ε_i. A) CRP concentrations C_c determined for each of our six experiments. The histogrammed distribution of each experiment-specific C_c value is shown. Fig. 4B in the main text plots the mean ± RMSD of these distributions. B) Distribution of values for the single CRP-RNAP interaction energy ε_i. The mean ± RMSD of this distribution is boxed and plotted in Fig. 4A. C) Each energy matrix shown as a heat plot in this paper is the optimal matrix found during Monte Carlo sampling. Panel C shows that optimal energy matrix elements are nearly identical to the mean matrix elements found through Monte Carlo sampling, confirming that optimal energy matrices are not outliers of the Monte-Carlo-sampled distributions. D,E) Plot of mean vs. RMSD values of CRP (D) and RNAP (E) energy matrix elements. The uncertainty in each matrix element (typically < 0.05 kcal/mol) is much less than the range of mean values (~2.5 kcal/mol), indicating that both matrices are very precisely determined. We do not currently understand why the relationship between these mean and RMSD values, especially for RNAP, is so precisely linear.

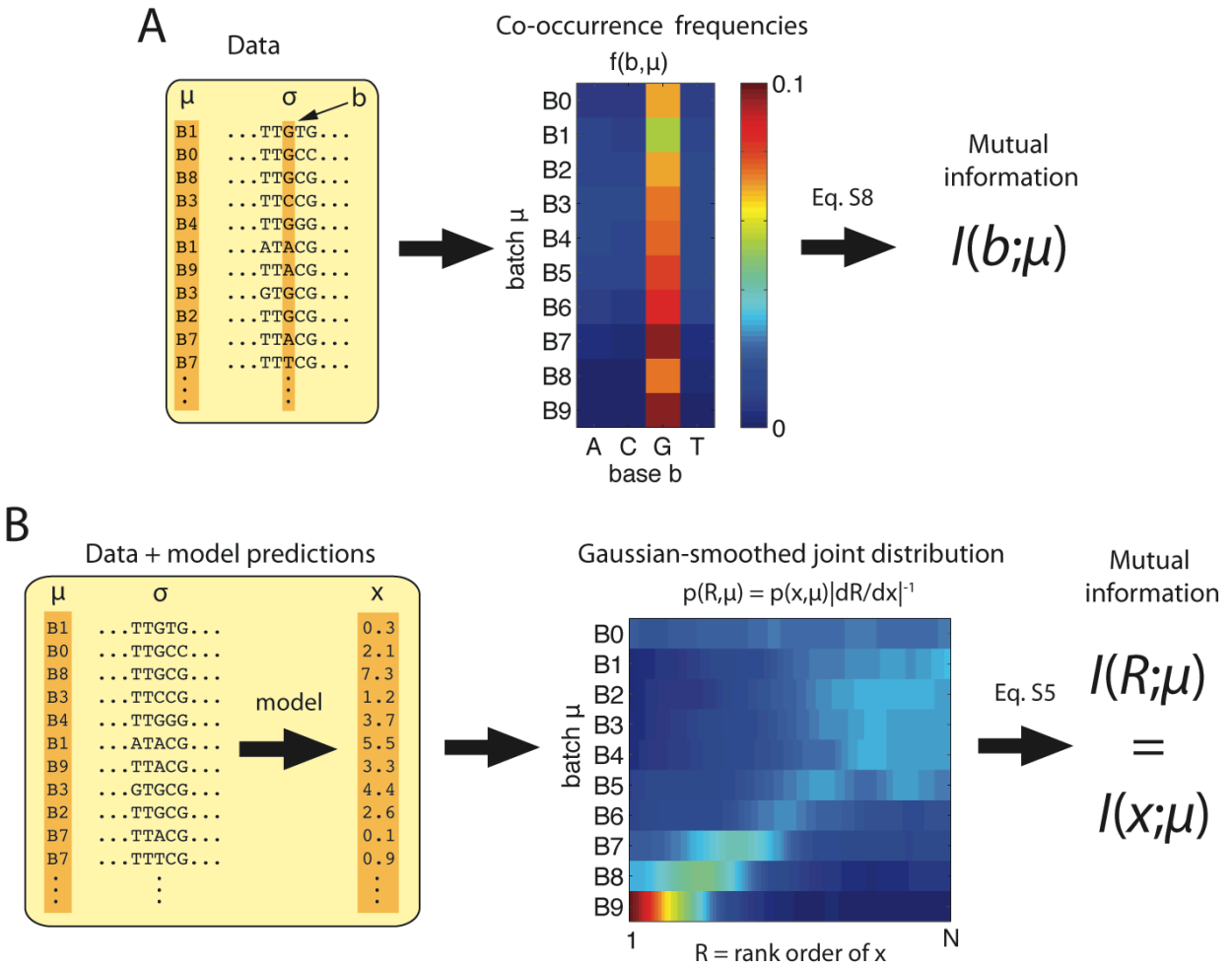

Figure S6. Computing mutual information. A) To compute $I(b;\mu)$ for information footprints, we first tallied the co-occurrence frequencies $f(b,\mu)$ of batches μ with the bases b at a specific nucleotide position. Eq. S15 was then used to compute the mutual information $I(b;\mu)$ from $f(b,\mu)$ using the finite-data correction of Treves and Panzeri, 1995. B) Computing predictive information $I(x;\mu)$ for model inference using Eq. 1. The model of interest was first used to assign a prediction x to each sequence σ. The corresponding rank order R of each prediction x was then computed. The resulting joint distribution, $p(R,\mu)$, was next regularized by blurring each data point with a Gaussian distribution in the R-direction. Finally, Eq. S5 was used to compute the mutual information between R and μ. This information is the same as $I(x;\mu)$, the predictive information of the model of interest.

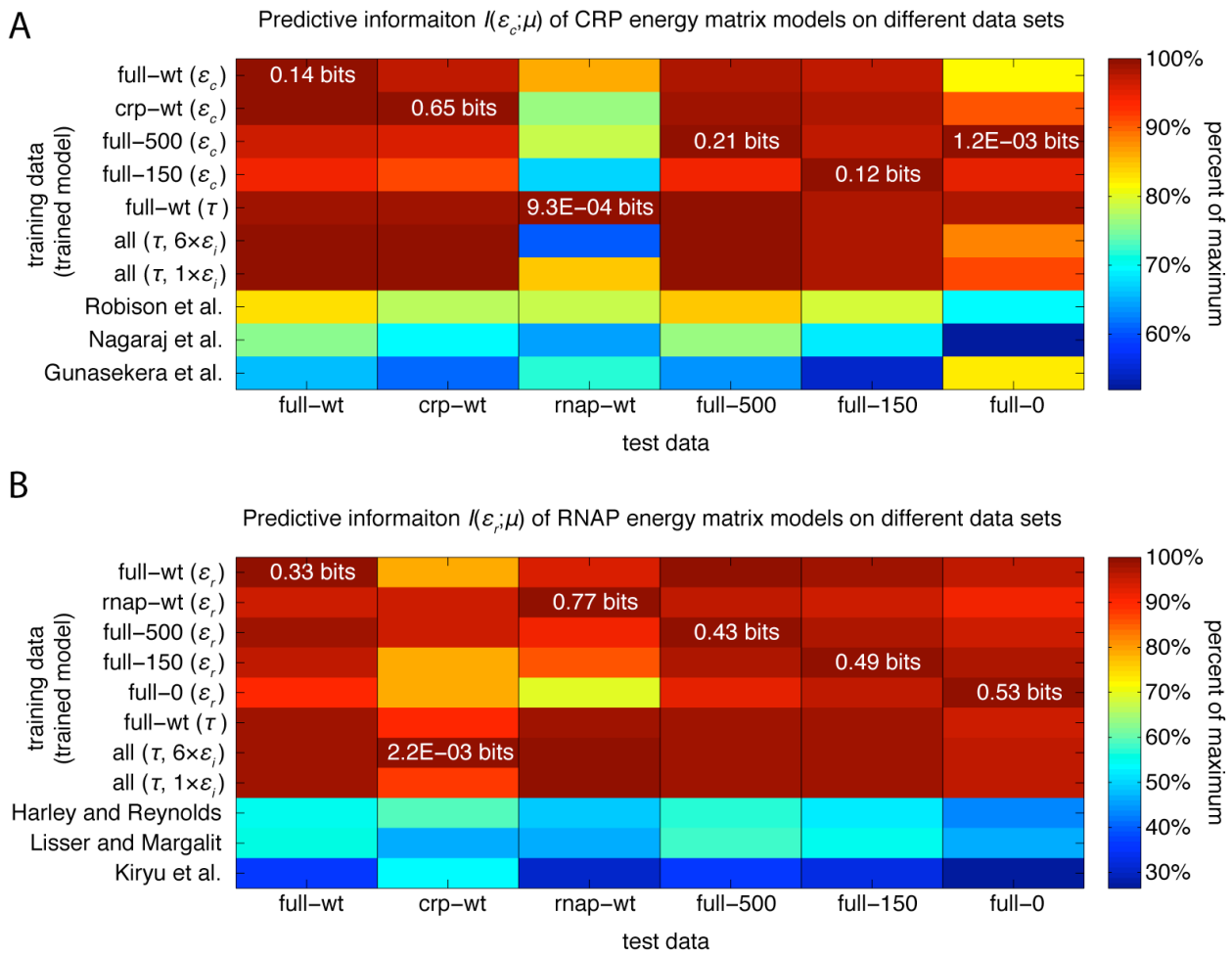

Figure S7. Performance of energy matrix models on different data sets. The relative predictive information values of A) 10 different CRP matrices or B) 11 different RNAP matrices are shown for each of the data sets from our six experiments. Each column corresponds to a given set of test data, i.e. data on which $I(\varepsilon_c;\mu)$ or $I(\varepsilon_r;\mu)$ was computed. In each column, the information value in white text indicates the maximum predictive information ($I(\varepsilon_c;\mu)$ or $I(\varepsilon_r;\mu)$) achieved by any of the tested models. Colors indicate the fraction of this maximum value achieved by each of the models tested. Models are annotated along the vertical axis according to the data set they were fit to. Note that some of the energy matrices tested here were inferred directly, while others were inferred as part of a model for τ. In addition, panels A and B each show the performance of three matrix models taken from the literature.